

Tech Breakthroughs · 2021 onwards

Research report · as of 2026-07-03 · AI · Bio · Energy · Space · Quantum

This report tracks 16 technology breakthroughs that have broken through from 2021 onwards — from transformers and reasoning models to GLP-1 agonists, CRISPR therapies, sub-threshold logical qubits, the Mechazilla booster catch, the collapse of battery costs below \$110/kWh, and commercial driverless robotaxis. For each breakthrough we identify the people behind it, the component *sub-ideas* that make it up, the cutting edge of the last 12–18 months, the rising organizations and individuals to watch, and what to look for next. All claims are footnoted to primary sources where possible. Where a claim could not be independently re-verified, we have flagged it.

Latest weekly highlights

Week ending 2026-07-03. Busy, well-sourced week led by frontier-model launches and unusually strong gap-filling across bio and the physical sciences. OpenAI opened a government-vetted limited preview of the GPT-5.6 family (Sol/Terra/Luna) and Anthropic shipped Claude Sonnet 5 at cut-rate pricing, while an LLM+Lean-4 loop (Claude Fable 5) produced the first machine-verified proof of the decade-old Farhi-Goldstone-Gutmann QAOA conjecture. Bio delivered Moderna's 5-year KEYNOTE-942 mRNA cancer-vaccine data, China's approval of the first commercial invasive BCI, and Grail's Galleri MCD trial missing its primary endpoint; physical-science items filled long-standing gaps (NASA CLPS lunar landers, SMRs, datacenter liquid cooling, direct-to-cell, PQC policy, quantum sensing). Three trending theses recurred (local open-weight coding agents, AI-monetization-beats-bubble data, and the frontier-access/monetization-window debate). Verification ran adversarially over 45 candidates (30 pass / 11 fix / 4 drop / 1 pre-tracked duplicate); entity reconciliation added 3 players; ~10 gap sub-themes flip to active.

- **OpenAI opens government-gated preview of GPT-5.6 Sol/Terra/Luna** — OpenAI began a limited preview of the GPT-5.6 family — Sol (flagship), Terra, and Luna — with Sol setting a new state of the art on Terminal-Bench 2.1 and priced \$5/\$30 per-M input/output tokens; broad release is gated by a US-government access process tied to a forthcoming cyber Executive Order framework.

- **Anthropic ships Claude Sonnet 5 at cut-rate introductory pricing** — Claude Sonnet 5 launched June 30 as the default for Free/Pro plans with introductory \$2/\$10 per-M input/output pricing (rising to \$3/\$15 after Aug 31); Anthropic reports gains on BrowseComp, OSWorld-Verified and Humanity's Last Exam and materially improved autonomous tool-use/agent reasoning over Sonnet 4.6. (Corrected: earlier draft's '1M context' and 'GDPval-AA' claims were not in the source and were removed.)

- **LLM + Lean-4 loop machine-verifies decade-old FGG QAOA conjecture** — Uri Kol, Maor Ben-Shahar, Kfir Sulimany and Dirk Englund (MIT) report the first machine-verified resolution of the Farhi-Goldstone-Gutmann conjecture (that depth-p QAOA on the ring of disagrees attains ratio $(2p+1)/(2p+2)$); the proof was found by Claude Fable 5 in an agentic feedback loop and certified end-to-end in Lean 4 (arXiv 2606.29687, submitted Jun 29 2026).

- **XDOF launches with \$70M to fix the robot teleoperation-data bottleneck** — XDOF launched with \$70M (Thrive Capital, Spark Capital, a16z, Lux Capital, WndrCo) and open-sourced ABC-130K, one of the largest open bimanual manipulation datasets (130K trajectories), targeting the teleoperation-data gap that constrains VLA/robot-foundation training.

- **Ambi Robotics + Pickle Robot fully automate trailer-unload-to-palletize** — Announced Jun 30, Pickle Robot's trailer-unloading arms feed Ambi's AmbiStack AI palletizer via conveyor for continuous scan-and-stack with no manual handoff at the dock door — a physical-AI logistics integration aimed at Fortune-500 retail/logistics.

- **ICRA 2026 Best Paper (Medical Robotics): RCM-constraint surgical control** — A TUM team (Li, Sadeghian, Haddadin et al.) won ICRA 2026's Best Paper Award in Medical Robotics for remote-center-of-motion constraint-consistent dynamic control of surgical robots — a standout in the surgical-robotics area the board doesn't yet track.

Executive summary

Breakthroughs at a glance: top named figures (org-type tagged), current state, and one rising organization to watch.

#	Breakthrough	Domain	Top people	Current state (1 line)	Watch this
01	Transformers / LLMs / coding agents	AI	Ashish Vaswani (Biz) Noam Shazeer (Biz)	Mixture of Experts (MoE) architecture has become the standard for frontier models, with DeepSeek-V3 activating only 5.5% of parameters per token, enabling efficient trillion-par...	Cursor (Anysphere)
02	RLHF + reasoning models (test-time compute scaling)	AI	Richard Sutton (Cross) Andrew Barto (Acad.)	DeepSeek-R1 demonstrates that pure reinforcement learning without supervised fine-tuning can achieve o1-level reasoning performance, published under MIT license with visible cha...	xAI Grok 5
03	Diffusion models (image + video generation)	AI	David Holz (Biz) Emad Mostaque (Biz)	Runway Gen-4.5 (December 2025) demonstrates advanced physics understanding including momentum, weight, fluid dynamics, and object interactions, with native audio generation inte...	Stability AI Cascade Model
04	AlphaFold + computational protein design	BIO	Demis Hassabis (Biz) John Jumper (Biz)	AlphaFold 3 (published May 2024 in Nature) predicts structures of proteins, nucleic acids, ligands, ions and modified residues with substantially improved accuracy for protein-l...	Chai Discovery
05	mRNA vaccines + LNP delivery	BIO	Katalin Karikó (Acad.) Drew Weissman (Acad.)	Moderna and Merck's mRNA-4157 personalized cancer vaccine combined with Keytruda sustained 49% reduction in recurrence/death risk in melanoma at 5-year follow-up (announced Janu...	Moderna/Merck cancer vaccine program
06	GLP-1 agonists	BIO	Jens Holst (Acad.) Mads Krogsgaard Thomsen (Biz)	Retatrutide (GIP/GLP-1/glucagon triple agonist) showed 28.7% weight loss and A1C reductions up to 2.0% in Phase 3 TRANSCEND-T2D-1 trial (announced March 2026).	Retatrutide clinical program (Eli Lilly)
07	CRISPR therapies (Casgevy + base/prime editing)	BIO	Feng Zhang (Acad.) David Liu (Acad.)	Casgevy (exagamglogene autotemcel) FDA-approved December 8, 2023, for sickle cell disease in patients ≥12 years; uses CRISPR/Cas9 to induce fetal hemoglobin, eliminating vaso-oc...	Prime Medicine (prime editing platform)
08	Quantum error correction (logical qubits below threshold)	QUANTUM	Hartmut Neven (Biz) John Preskill (Acad.)	Google Willow (December 2024): 105-qubit processor achieved surface code error suppression below threshold with distance-7 code at 0.143% per-cycle logical error rate; first dem...	PsiQuantum
09	Reusable rockets (Falcon 9 + Starship catch)	SPACE	Elon Musk (Biz) Gwynne Shotwell (Biz)	Falcon 9 Booster Reusability Record (May 2026): 384+ booster reflights with 100% success rate; single booster B1062 reached 22 flights by early 2026; turnaround as short as 3 we...	Blue Origin New Glenn
10	Renewables + battery cost collapse	ENERGY	Jay Whitacre (Cross) Tejas Patel (Biz)	BloombergNEF Battery Price Collapse (June 2025): Lithium-ion pack prices fell 8% to record low of \$108/kWh; stationary storage at \$70/kWh (45% drop YoY); BEV packs at \$99/kWh (2...	BYD Sodium-Ion
11	Self-driving cars / robotaxis	AI	Tekedra Mawakana (Biz) Dmitri Dolgov (Biz)	Waymo One scaled paid driverless rides from a single Phoenix geofence (2020) to commercial operation across Phoenix, San Francisco, Los Angeles, Austin, and Atlanta, surpassing ...	Wayve

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12	Humanoid robotics	ROBOTICS	Sergey Levine (Biz) Jitendra Malik (Acad.)	Figure's Helix 02 (Jan 2026) shows unified full-body autonomy from a single VLA — dishwasher loading, bottle uncapping, pill extraction, syringe dispensing — without task-specif...	Figure AI
13	Fusion energy	ENERGY	Omar Hurricane (Gov) Bob Mumgaard (Biz)	NIF achieved 10+ ignition shots with a peak yield of 8.6 MJ (April 2025) — establishing repeatable net energy gain and a target-gain trend above 4:1 for inertial confinement fus...	Commonwealth Fusion Systems (SPARC)
14	Brain-computer interfaces	BIO	Leigh Hochberg (Cross) Edward Chang (Acad.)	Neuralink's PRIME trial has implanted 5+ patients across multiple sites by mid-2025; the second patient ('Alex') achieved CAD control and gaming within days, and Blindsight (vis...	Neuralink Blindsight
15	AI accelerators / advanced compute	HARDWARE	Jensen Huang (Biz) Andrew Feldman (Biz)	NVIDIA Rubin GPUs ship in 2H 2026 with ~3× Blackwell performance and 1.2 EFLOPS FP8 training capability; Rubin Ultra follows in 2027.	Tenstorrent
16	AI dev ecosystem & agent primitives (MCP, skills, agents, orchestration)	AI	David Soria Parra (Biz) Justin Spahr-Summers (Biz)	MCP went from an Anthropic experiment (announced Nov 25, 2024) to a cross-industry standard; in Dec 2025 Anthropic donated MCP to the Agentic AI Foundation, a Linux Foundation d...	Conductor

Transformers / LLMs / coding agents

Impact so far. Transformers and LLMs have enabled agentic AI systems that autonomously solve software engineering tasks, with frontier models now matching or exceeding professional developer capabilities on real-world codebase tasks.

Watch for next. Watch for scaling reasoning with test-time compute to surpass human performance on complex multi-step planning; emergence of true agentic systems that coordinate multiple specialized coding agents in parallel; and development of interpretable reasoning traces that enable verification and debugging of AI agent decisions.

a) People behind it

Name	Role / Affiliation	Type	Prizes
Ashish Vaswani ^[1]	Co-founder and CEO, Essential AI; Co-author of 'Attention Is All You Need' Essential AI	Biz	—
Noam Shazeer ^[2]	VP Engineering, Gemini Co-lead Google DeepMind	Biz	—
Dario Amodei ^[3]	CEO Anthropic	Biz	—
Ilya Sutskever ^[4]	CEO Safe Superintelligence Inc. (SSI)	Biz	—
Jeff Dean ^[5]	Chief Scientist, Co-lead Gemini Google DeepMind	Biz	—
Yann LeCun ^[6]	Founder and Chief Scientist Advanced Machine Intelligence Labs (AMI Labs); Jacob T. Schwartz Chaired Professor, NYU Courant Institute	Cross	★ ACM A.M. Turing Award 2018
John Jumper ^[7]	Joining Anthropic (research) after ~9 years leading AlphaFold at Google DeepMind Anthropic	Biz	★ 2024 Nobel Prize in Chemistry (AlphaFold)

Sub-ideas inside this breakthrough

Sub-idea	Description	Examples
Coding agents ^[8]	Agentic LLMs that autonomously edit, run, and debug code have shifted programming from write-then-run to delegate-then-review — Cursor, Devin, Claude Code, and GitHub Copilot Workspace are now standard tooling for many engineering teams.	Cursor · Devin (Cognition) · Claude Code · GitHub Copilot Workspace · Aider
Mixture-of-Experts (MoE) ^[9]	Conditional computation that activates only a small fraction of parameters per token (DeepSeek-V3 ~5.5%) lets trillion-parameter models stay tractable to serve, decoupling parameter count from per-token cost.	DeepSeek-V3 · Mixtral 8x22B · Qwen3-MoE · Grok-1
Long-context windows ^[10]	Production context lengths exploded from 4K (GPT-3.5) to 1M+ tokens (Gemini, Claude Sonnet 4, Qwen2.5-1M), enabling whole-codebase reasoning, long-document analysis, and persistent agent memory.	Gemini 2.5 Pro 2M · Claude Sonnet 4 1M · GPT-4.1 1M · Qwen2.5-1M
Multimodal frontier models ^[11]	Native multimodal training (vision + audio + text + code in one model) replaced bolt-on adapters and is now the default for frontier systems, supporting voice agents, screen-reading, and embodied use cases.	GPT-4o · Gemini 1.5/2.5 · Claude 3.5/4 Vision · Llama 3.2 Vision
Open-weight catch-up ^[12]	Open-weight models (Llama, Qwen, DeepSeek) closed most of the gap with closed-frontier systems, with DeepSeek-V3 and R1 published with permissive MIT-style licenses, restructuring the geopolitics of AI compute.	DeepSeek-V3 / R1 · Llama 3.x · Qwen2.5 / Qwen3 · Mistral Large

Sub-idea	Description	Examples
Model Context Protocol (MCP) ^[13]	Anthropic's open MCP standard (Nov 2024) became the lingua franca for connecting LLMs to external tools and data — adopted across Claude, OpenAI, and most agent frameworks, replacing one-off plugin systems with a single client/server protocol.	Anthropic MCP · MCP servers (Slack, GitHub, Linear, Notion) · Claude Desktop · OpenAI MCP support
AWS MCP Server (managed) ^[14]	Managed MCP server providing secure, auditable agent access to AWS services with file uploads, long-running operations, sandboxed Python, IAM guardrails, and CloudTrail logging.	Agents can call any AWS API via single tool · Sandboxed Python execution for multi-step operations · 40+ pre-built skills for IaC, storage, analytics, serverless
GPT-Realtime-2 voice stack ^[15]	Native real-time voice agent stack with GPT-5-class reasoning, 70+ language live translation, and streaming Whisper transcription, exposed via the OpenAI API.	Real-time voice agent with reasoning + tool use · Live multilingual translation in 70+ languages · Streaming low-latency transcription
OpenAI GPT-5.5-Cyber (Trusted Access for Cyber) ^[16]	May 7, 2026 — Domain-specialized variant of GPT-5.5 for vetted cyber defenders, launched via Trusted Access for Cyber identity tier. Permits bug-hunting, malware analysis, reverse engineering; blocks credential theft and malware writing. Cisco and CrowdStrike are launch partners; Axios sources peg vuln-finding capability ~ Anthropic Mythos Preview.	Vetted SOC and red-team teams at Cisco / CrowdStrike-tier orgs gain access to the more-permissive variant · Phishing-resistant auth required from June 1, 2026 for the highest access tier
OS-native agentic AI layer ^[17]	A device-level agent that understands screen context, moves across apps and the browser, and completes multi-step tasks with the user in the loop — turning the smartphone OS itself into the agent surface.	Google Gemini Intelligence for Android (May 12, 2026) · Gemini in Chrome (June 2026) · Gboard Rambler dictation agent
Streaming / predictive RAG for voice agents ^[18]	Decouple retrieval from generation so the vector-DB round-trip never blocks speech: a background agent anticipates likely queries and pre-fetches into a fast cache while the speaking agent reads only from cache (or retrieves in parallel with user speech).	VoiceAgentRAG dual-agent 'Slow Thinker'/'Fast Talker' memory router (arXiv 2603.02206) · Stream RAG streaming tool-call prediction during user speech (arXiv 2510.02044)
IndexShare shared sparse-attention indexer ^[19]	A sparse-attention efficiency method that reuses a single indexer across groups of four attention layers, reducing per-token FLOPs by 2.9x at 1M context, demonstrated at frontier quality in GLM-5.2.	GLM-5.2 (Z.ai, June 2026): 2.9x per-token FLOP reduction at 1M context via IndexShare

b) Current research frontier (last 12–18 months)

1. Mixture of Experts (MoE) architecture has become the standard for frontier models, with DeepSeek-V3 activating only 5.5% of parameters per token, enabling efficient trillion-parameter deployment.^[20]
2. Context length scaling to 1M tokens is now production-ready with Gemini 2.5 Pro (2M tokens), Claude Sonnet 4 (1M tokens), and open-source Qwen2.5-1M.^[21]
3. Claude Opus 4.7 and GPT-5.5 represent 2026 frontier capabilities with improved instruction persistence and multi-step agentic orchestration.^[22]
4. Claude Mythos Preview achieves 93.9% on SWE-bench Verified, though only 45.9% on the harder contamination-free SWE-bench Pro, revealing data contamination in benchmark evaluation.^[23]
5. Diffusion transformers with patch-based representations have emerged as scalable architecture for both language and vision domains, with Next-Latent Prediction extending self-supervised learning to latent space.^[24]
6. GPT-5.5 Instant produces 52.5% fewer hallucinated claims than GPT-5.3 Instant on high-stakes prompts in medicine, law, and finance (OpenAI release notes, May 5 2026).^[25]
7. May 6, 2026 — Anthropic secures all of SpaceX Colossus 1 (~300 MW / 220K+ GPUs within the month), doubles Claude Code rate limits, raises Opus API limits, and expresses interest in multi-GW orbital compute with SpaceX.^[26]
8. Gemini 3.5 Flash is a frontier-rivaling small model optimized for agentic coding, GA in Antigravity 2.0 and the Gemini API as of May 19, 2026.^[27]
9. Claude Opus 4.8 released (May 28, 2026): 69.2% agentic coding, 84% computer-use on Online-Mind2Web, new dynamic workflows for Claude Code.^[28]
10. Karpathy joined Anthropic pretraining team May 2026 to lead a 'use Claude to accelerate pretraining' effort.^[29]
11. MiniMax M3 (June 1, 2026): open-weight model with 1M-token context via MiniMax Sparse Attention, 59.0% SWE-Bench Pro (beats GPT-5.5 and Gemini 3.1 Pro, approaches Opus 4.7), native multimodality and computer use; weights+report within 10 days.^[30]

- 12.** Claude Fable 5, a new 'Mythos-class' tier sitting above Opus, is now generally available and described by Anthropic as SOTA on nearly all tested AI-capability benchmarks, with classifier fallback to Opus 4.8 on cyber/bio/distillation queries.^[31]
- 13.** Open-weight frontier coding/agentic model GLM-5.2 (744B MoE, MIT, 1M context) becomes first open model >80% on Terminal-Bench 2.1 (81.0) and 62.1 SWE-bench Pro, beating GPT-5.5 at ~1/6 cost.^[19]
- 14.** Two June 2026 Nature papers show agentic AI doing autonomous, multi-step, end-to-end clinical management at or above board-certified-physician performance in simulated settings — MIRA (ED, 87.8% vs 78.1% dx accuracy) and Google's AMIE (longitudinal outpatient, non-inferior to PCPs).^[32]
- 15.** OpenAI limited-preview of GPT-5.6 Sol/Terra/Luna; Sol new SOTA on Terminal-Bench 2.1, \$5/\$30 per-M tokens; broad release gated by a US-government access process tied to a forthcoming cyber EO.^[33]
- 16.** Claude Sonnet 5 (Jun 30) default for Free/Pro, intro \$2/\$10 per-M pricing (→\$3/\$15 after Aug 31); gains on BrowseComp, OSWorld-Verified, Humanity's Last Exam and improved agentic tool-use vs Sonnet 4.6.^[34]

c) Who could be next

Cursor (Anysphere) [STARTUP]^[35] — Anysphere Inc. · Series D, \$29.3B valuation (2026)

Fastest-growing AI coding IDE, reaching \$29.3B valuation with \$2B+ ARR by early 2026, supporting parallel multi-agent coding.

Cognition Labs (Devin) [STARTUP]^[36] — Cognition Labs · Series C+, \$400M at \$10.2B valuation; targeting \$25B (2026)

Autonomous AI software engineer scaling ARR from \$1M to \$73M in 9 months; acquired Windsurf; targeting \$25B valuation in April 2026.

Safe Superintelligence Inc. (SSI) [STARTUP]^[37] — SSI · Series B, \$2B at \$32B valuation (April 2025)

Sutskever's new lab focused on safety-first AGI path with \$3B+ raised and \$32B valuation; significant backing from Alphabet and Nvidia.

Advanced Machine Intelligence Labs (AMI Labs) [STARTUP]^[6] — AMI Labs · Series A, \$1.03B at \$3.5B (March 2026)

Yann LeCun's new venture focused on world models and open-source development; raised \$1.03B at \$3.5B pre-money valuation in March 2026.

Anthropic enterprise-AI JV [STARTUP]^[38] — Anthropic / Blackstone / Hellman & Friedman / Goldman Sachs · \$1.5B initial capitalization; backed by Blackstone, H&F, Goldman Sachs

First services-led commercialization vehicle for a frontier-lab model; converts mid-market deployments from pure API consumption to verticalized AI services.

Z.ai (Zhipu AI) [LAB]^[19] — Z.ai / Zhipu AI · Well-funded Chinese AI 'tiger' (multi-billion valuation; public-listing track)

Leading Chinese open-weight frontier lab (GLM series); GLM-5.2 is the first open model >80% Terminal-Bench 2.1 and a credible MIT-licensed competitor to closed frontier coding models at ~1/6 cost.

Flags / unverified

■ SWE-bench Verified benchmark scores may be inflated due to data contamination; SWE-bench Pro (contamination-free) shows significantly lower performance

■ Meta's Llama benchmark results were reportedly fudged according to departing chief scientist Yann LeCun

■ No verified information found on Tom Brown's current 2026 affiliation beyond his core role at Anthropic

■ Updated Anthropic Series H funding closed: new value = \$65B raised at \$965B post-money (May 28, 2026); leads Altimeter, Dragoner, Greenoaks, Sequoia; \$47B run-rate revenue

RLHF + reasoning models (test-time compute scaling)

Impact so far. Reasoning models with test-time compute scaling have fundamentally shifted AI capability distribution from training time to inference time, enabling smaller models to solve previously intractable problems in mathematics, coding, and science while achieving cost-competitive inference.

Watch for next. Monitor scaling laws for reasoning beyond current limits; emergence of multi-step verifiable reasoning that enables formal proof verification; and development of interpretable chain-of-thought that reveals genuine reasoning versus pattern matching in reasoning model outputs.

a) People behind it

Name	Role / Affiliation	Type	Prizes
Richard Sutton ^[39]	Professor, University of Alberta; Senior Scientist, DeepMind University of Alberta / DeepMind	Cross	★ ACM A.M. Turing Award 2024 (announced 2025)
Andrew Barto ^[39]	Emeritus Professor University of Massachusetts Amherst	Acad.	★ ACM A.M. Turing Award 2024 (announced 2025)
Ilya Sutskever ^[37]	CEO Safe Superintelligence Inc.	Biz	—
Dario Amodei ^[3]	CEO Anthropic	Biz	—
Demis Hassabis ^[40]	CEO Google DeepMind	Biz	—
DeepSeek Research Team ^[12]	Lead researchers on DeepSeek-R1 DeepSeek-AI	Biz	—

Sub-ideas inside this breakthrough

Sub-idea	Description	Examples
Test-time compute scaling ^[41]	Pioneered by OpenAI o1/o3, scaling inference-time tokens (often 10–100x more thinking) yields steeper accuracy gains than equivalent training-time scaling on math, code, and science benchmarks.	OpenAI o1 / o3 · Gemini 2.5 Thinking · Claude reasoning
Pure-RL reasoning (R1-Zero recipe) ^[12]	DeepSeek-R1 showed that reasoning behaviors emerge from pure reinforcement learning without any supervised fine-tuning, just by rewarding correct final answers — collapsing what was thought to be a complex pipeline.	DeepSeek-R1 · DeepSeek-R1-Zero · Qwen-QwQ
RLHF → RLAIF → Constitutional AI ^[42]	Human preference data, the bottleneck of early RLHF, has been progressively replaced by AI-generated feedback against written principles (Anthropic’s Constitutional AI) and verifiable reward functions (RLVR).	Anthropic Constitutional AI · RLAIF · RLVR
Reasoning distillation ^[43]	Long chains of thought from large reasoning models can be distilled into much smaller students (DeepSeek-R1 → Qwen-7B), bringing o1-class behavior into the open-weight ecosystem at a tiny fraction of the cost.	R1-distill-Qwen-32B · R1-distill-Llama-70B · Sky-T1
Agentic reasoning + tool use ^[44]	Reasoning models now interleave planning steps with tool calls (browsers, code execution, retrieval), turning a chat model into a research assistant that runs a multi-step plan against real systems.	OpenAI Deep Research · Claude tool use · Gemini DeepThink
Step-wise rubric rewards (SRaR) ^[45]	RLVR variant that decomposes rubric-based supervision onto individual reasoning steps and normalizes across rollouts to avoid uniform credit assignment and reward hacking.	SRaR (arXiv 2605.17291, May 17, 2026)

Sub-idea	Description	Examples
Token-level credit assignment for RLVR ^[46]	Per-token discriminative credit signal layered on top of outcome-verifiable RL, giving denser gradient signal than response-level rewards.	DeITA (arXiv 2605.21467, May 20, 2026)
Equilibrium Reasoners (latent-attractor TTC) ^[47]	Iterative latent-attractor reasoning enabling deep test-time-compute scaling without external verifiers; pushes Sudoku-Extreme accuracy from 2.6% to 99%+ on unrolling equivalent to ~40k layers.	Sudoku-Extreme 2.6% -> 99%+ accuracy by unrolling latent attractors
Long-horizon autonomous agentic reasoning ^[30]	Evaluating reasoning models by their ability to run unattended for many hours through plan-execute-diagnose-iterate loops on real research and systems tasks, measured by submissions/tool-calls/commits and end-state performance rather than single-pass accuracy.	MiniMax M3 reproducing an ICLR 2025 paper autonomously over ~12h (18 commits, 23 figures) · MiniMax M3 optimizing an FP8 GEMM kernel on NVIDIA Hopper over ~24h (147 submissions, 1,959 tool calls), 7.6%->71.3% peak utilization
Autonomous scientific hypothesis generation ^[31]	Frontier reasoning models running long autonomous research loops (protein design, genomics) that propose and validate novel hypotheses at or above expert level — test-time/agentic compute applied to open-ended science.	Mythos 5 preferred ~80% in blinded molecular-biology hypothesis comparisons · Mythos 5 trained a single-cell genomics model that beat a Science-published model while ~100x smaller · Mythos 5 accelerated internal protein/drug design ~10x

b) Current research frontier (last 12–18 months)

- DeepSeek-R1 demonstrates that pure reinforcement learning without supervised fine-tuning can achieve o1-level reasoning performance, published under MIT license with visible chain-of-thought reasoning.^[12]
- Test-time compute scaling laws show that allocating 2-10x more inference tokens for reasoning can match or exceed accuracy of larger models, with smaller models more compute-efficient than scaling parameters.^[41]
- OpenAI o3 achieves 45.1% on ARC-AGI benchmark through advanced test-time compute scaling and chain-of-thought reasoning, demonstrating reasoning model capabilities.^[48]
- RLHF methods in 2025 now employ verifiable rewards via direct comparison scoring (RLVR) and online iterative feedback collection, achieving state-of-the-art on AlpacaEval-2 and Arena-Hard benchmarks.^[49]
- Anthropic's constitutional AI systems with classifier jailbreak detection withstood 3,000+ hours of expert red-teaming with minimal universal jailbreaks discovered.^[42]
- Sparse policy selection at entropy-gated decision points matches or exceeds full-RL reasoning at ~1000x lower training cost (ReasonMaxxer, arXiv 2605.06241).^[50]
- RL compute for long-horizon reasoning follows a power law in reasoning depth; scaling exponent monotonically increases with logical expressiveness (1.04 -> 2.60) (ScaleLogic, arXiv 2605.06638).^[51]
- Search-driven reward optimization improves reasoning on GSM8K without retraining the base model (arXiv 2605.02073).^[52]
- TraceLift multiplies rubric-based Reasoning Reward Model scores by measured uplift on frozen executors, crediting reasoning traces that are useful to downstream consumers (arXiv 2605.03862).^[53]
- May 7, 2026 — Anthropic publishes Natural Language Autoencoders (NLAs); RL-trained verbalizer/reconstructor pair turns activations into readable explanations; lifts hidden-motivation auditing from <3% to 12-15% and exposes evaluation-awareness on 16-26% of test transcripts vs <1% of real usage; used in Mythos Preview and Opus 4.6 alignment audits.^[54]
- Self-supervised world-model learning now has a machine-checkable (Lean 4) identifiability guarantee: LeJEPA provably recovers a world's true latent variables iff the latents are Gaussian and evolve under stationary additive-noise dynamics (arXiv 2605.26379, May 2026).^[55]
- Pairing LLMs with the Lean 4 proof assistant turns AI math/code outputs from plausible-but-unverified into machine-checked-correct; AxiomProver autonomously generated Lean/Mathlib-verified proofs (Axiom Math raised a \$200M Series A led by Menlo Ventures). [BACKFILL, arXiv Feb 2026]^[56]
- Karpathy released 'autoresearch' (Mar 7, 2026, MIT, ~66k+ GitHub stars): a minimal single-GPU nanochat harness where a bring-your-own coding agent (Claude/Codex) autonomously edits train.py, runs fixed 5-min experiments, and keeps/discards on val_bpb — the concrete milestone crystallizing the AI-doing-AI-research thesis. Notable runs: Shopify +19% (37 experiments), Karpathy 700-experiment run (time-to-GPT-2 2.02h→1.80h).^[57]

14. For long-horizon agentic coding, standard test-time compute scaling does not transfer cleanly; the central bottleneck is representing and reusing prior agent experience rather than adding inference compute.^[58]
15. Google Big Sleep (DeepMind + Project Zero) found CVE-2025-6965 (SQLite) before in-the-wild exploitation — first AI agent to foil a real-world exploit attempt.^[59]
16. DARPA AlxCC final (DEF CON 33): autonomous systems scanned 54M LoC, found 86% of synthetic vulns / patched 68%, and found 18 real zero-days at ~\$152/task; Team Atlanta won.^[60]
17. CyberGym (UC Berkeley/Dawn Song, arXiv:2506.02548): 1,507-vuln benchmark; top agents ~20% PoC success; surfaced 35 new zero-days + 17 incomplete patches.^[61]
18. Anthropic disrupted the first reported AI-orchestrated cyber-espionage campaign (GTG-1002 jailbroke Claude Code; AI ran 80-90% of recon/exploit/exfil against ~30 targets).^[62]
19. OpenAI Aardvark (GPT-5 agentic security researcher): 92% recall on golden repos, found bugs that received 10 CVE IDs; became Codex Security.^[63]
20. Microsoft MatterGen (Nature, Jan 2025): diffusion generative model for inorganic materials; >2x more stable/unique/new outputs, >10x closer to DFT minimum; synthesized TaCr2O6 within 20% of target.^[64]
21. Meta FAIR OMat24 (arXiv:2410.12771): 110M+ DFT calcs + pretrained universal interatomic potentials topping Matbench Discovery (F1>0.9, ~20 meV/atom).^[65]
22. Microsoft Aurora (Nature, May 2025): 1.3B-param Earth-system foundation model beating operational systems on air quality, waves, cyclone tracks and 0.1° weather; >91% vs GraphCast.^[66]
23. DeepMind GenCast (Nature, Dec 2024): 0.25° diffusion ensemble beats ECMWF ENS on 97.2% of 1,320 targets to 15 days; ~8 min/forecast on one TPU v5; open-sourced.^[67]
24. DeepMind WeatherNext 2 (Nov 2025): FGN model, 8x faster, hundreds of scenarios in <1 min on one TPU, beats prior WeatherNext on 99.9% of variables; shipped to Search/Gemini/Maps.^[68]
25. First machine-verified proof of the decade-old FGG QAOA approximation-ratio conjecture, found by Claude Fable 5 and certified in Lean 4 (Kol/Ben-Shahar/Sulimany/Englund, MIT; arXiv 2606.29687).^[69]

c) Who could be next

xAI Grok 5 [STARTUP]^[70] — xAI / SpaceX · Private; xAI acquired by SpaceX in February 2026

Grok 4 (July 2025) implements reasoning similar to o3-mini and DeepSeek-R1; Grok 5 under development with target AGI-level claims for 2025-2026 timeframe.

DeepSeek Research [STARTUP]^[43] — DeepSeek-AI · Private; backed by Chinese investors

Leading open-source reasoning model work with DeepSeek-R1 (MIT license); R1 distilled models on Qwen achieving near-parity with proprietary reasoning models.

Anthropic Scaling Team [LAB]^[71] — Anthropic · Series C funding; \$380B company valuation (Feb 2026)

Leading research on constitutional AI, verifiable rewards, and test-time compute for Claude; achieving frontier performance on reasoning benchmarks.

OpenAI o-series Development [LAB]^[72] — OpenAI · Private; multi-billion funding rounds

Pioneered test-time compute scaling with o1 and o3; o3 sets ARC-AGI benchmark records and defines the frontier of inference-time reasoning.

XBOW [STARTUP]^[73] — XBOW USA Inc. · Series C \$120M at \$1B+ (Mar 2026), led by DFJ Growth & Northzone; prior \$75M Series B (Altimeter, Sequoia, NFDG)

Autonomous offensive-security agent; first to top HackerOne's US leaderboard; deployed at Fortune 500s. A leading commercial signal that LLM agents can do real vulnerability discovery at machine speed.

RunSybil [STARTUP]^[74] — RunSybil · \$40M (Mar 2026) led by Khosla Ventures; S32, Anthology Fund (Anthropic/Menlo), Conviction, Elad Gil

Autonomous black-box pentesting of live applications (vs static code scanning); founded by OpenAI's first security hire. Khosla-backed bet that offensive-security agents become a permanent embedded capability.

Periodic Labs [STARTUP]^[75] — Periodic Labs · ~\$300M seed (Oct 2025), led by a16z (Felicis first check); DST, NVentures, Accel; angels Bezos, Schmidt, Gil, Jeff Dean

Pairs LLMs + simulation + robotic synthesis as an AI-automated materials lab; founded by GNoME co-author Cubuk and ex-OpenAI post-training lead Fedus. One of the most prominent bets that scientific FMs + autonomous labs can discover new materials.

Lila Sciences [STARTUP]^[76] — Lila Sciences (Flagship Pioneering) · \$200M seed (Mar 2025), led by Flagship Pioneering; General Catalyst, March Capital, ARK, ADIA

Building scientific-superintelligence + autonomous 'AI Science Factories' spanning materials/chemistry/life science; Flagship-incubated with George Church as chief scientist. A flagship bet on FM-driven autonomous experimentation.

Flags / unverified

■ No confirmed public arxiv paper on OpenAI o3 architecture or training methods as of May 2026

■ DeepSeek-R1 claimed performance parity with o1, but independent verification limited to December 2024 release; newer o-series improvements not directly compared

■ Test-time compute scaling laws may plateau at certain thresholds; optimal inference compute allocation strategy remains active research area

Diffusion models (image + video generation)

Impact so far. Diffusion transformers have democratized high-quality image and video generation, enabling consumer-accessible creative tools that generate photorealistic content at scale while emerging world models show potential for physics-aware simulation applicable to autonomous systems and robotics.

Watch for next. Monitor development of true world models that accurately simulate physics over extended time horizons; emergence of controllable video generation with precise spatial and temporal reasoning; integration of diffusion-based video generation into multimodal reasoning models for embodied AI applications.

a) People behind it

Name	Role / Affiliation	Type	Prizes
David Holz ^[77]	Founder and CEO Midjourney	Biz	—
Emad Mostaque ^[78]	Founder (formerly CEO) Stability AI	Biz	—
Oriol Vinyals ^[79]	VP Research, Gemini Co-lead Google DeepMind	Biz	—
Yann LeCun ^[6]	Founder, Advanced Machine Intelligence Labs AMI Labs; NYU Courant Institute	Cross	★ ACM A.M. Turing Award 2018
Tim Brooks ^[80]	Researcher (Sora team) OpenAI	Biz	—
Runway Research Team ^[81]	Lead researchers on Gen-4.5 and GWM-1 Runway	Biz	—

Sub-ideas inside this breakthrough

Sub-idea	Description	Examples
Latent diffusion ^[82]	Running diffusion in a compressed latent space rather than pixel space (Stable Diffusion, 2022) cut compute 50–100x and made image generation runnable on a laptop GPU, kicking off the consumer wave.	Stable Diffusion 1.x/3 · SDXL · Stable Cascade
Diffusion transformers (DiT) ^[83]	Replacing the U-Net with a transformer backbone (DiT, Peebles & Xie 2023) gave diffusion the same scaling properties as LLMs and is the architecture underneath Sora, Stable Diffusion 3, and Veo.	Sora · Stable Diffusion 3 · Veo · Imagen 3
Long-form video generation ^[84]	Sora, Veo, Runway Gen-4, and Kling moved video diffusion from 4-second clips to coherent minute-long shots with stable identities, camera control, and physics, opening up real production use.	OpenAI Sora · Google Veo 3 · Runway Gen-4.5 · Kuaishou Kling
World models ^[85]	Video diffusion is being repurposed as a learned simulator — Genie, GWM-1, and Cosmos generate playable, physics-aware environments for robotics and embodied AI rather than passive video.	DeepMind Genie 2 · Runway GWM-1 · NVIDIA Cosmos · World Labs
One-step / real-time generation ^[86]	Distillation techniques (consistency models, flow matching, adversarial diffusion) collapsed image generation from 50+ steps to 1–4, enabling real-time interactive generation in design and gaming tools.	Latent Consistency Models · SDXL Turbo · Adversarial Diffusion Distillation
World-grounded omni video generation ^[87]	Single model jointly trained for reasoning and multimodal generation, producing video that is grounded in real-world knowledge from text/image/audio/video inputs.	Gemini Omni Flash (Google, May 2026)

b) Current research frontier (last 12–18 months)

01. Runway Gen-4.5 (December 2025) demonstrates advanced physics understanding including momentum, weight, fluid dynamics, and object interactions, with native audio generation integrated.^[81]
02. Runway's GWM-1 world model enables frame-by-frame physics simulation with user-defined world parameters, representing shift from video generation to physics-aware simulation.^[85]
03. Sora 2 (September 2025) released with improved consistency and control, but OpenAI announced April 2026 shutdown and September 2026 API discontinuation, shifting to GPT-native video generation.^[88]
04. Google Imagen 3 (August 2024) surpasses DALL-E 3 and Stable Diffusion in photorealism and text rendering, achieving better prompt-image alignment across automated and human evaluations.^[89]
05. Midjourney V7 (April 2025, default June 2025) features 12B-parameter multimodal transformer with 20-30% faster rendering and improved hand/object coherence over V6.^[90]
06. NVIDIA Cosmos 3 (May 31, 2026): open Mixture-of-Transformers world-model family unifying physical reasoning (autoregressive VLM tower) and diffusion-based world/action generation; Nano (16B) and Super (64B) checkpoints, open-source SOTA on R-Bench, leading Artificial Analysis text-to-image and image-to-video.^[91]
07. Fused INT8 Triton kernel realizes native integer compute for diffusion transformers, enabling 1024px single-RTX-3090 generation (~9.5% end-to-end vs FP8; consumer-Ampere only). arXiv 2606.14598.^[92]
08. Ideogram 4.0 (Jun 3, 2026): open-weight 9.3B single-stream diffusion-transformer text-to-image foundation model (commercial license), positioned as closing the proprietary-vs-open image-model gap.^[93]
09. Google's DiffusionGemma (26B-A4B MoE, Apache 2.0, 2026-06-10) is an open-weight discrete-diffusion language model that generates 256-token canvases in parallel for up to 4x faster decoding (1000+ tok/s on one H100) with bidirectional self-correction.^[94]

c) Who could be next

Stability AI Cascade Model [LAB]^[95] — Stability AI · Venture-backed; facing financial difficulties since 2024

Stable Cascade achieves 2x faster inference than SDXL with 42x compression factor, enabling efficient fine-tuning (16x cost reduction) and edge deployment.

Open-Sora 2.0 [LAB]^[96] — HPAI Tech · Academic/open-source; community-driven

Commercial-level video generation trained for only \$200k; reduced performance gap with OpenAI Sora from 4.52% to 0.69%, democratizing video generation.

Runway Media Creation Platform [STARTUP]^[81] — Runway · Series C, \$500M+ valuation estimates

Leading commercial video generation with Gen-4.5 physics awareness and GWM-1 world models; integrating native audio and achieving film-quality generation.

Midjourney Next-Gen Development [STARTUP]^[97] — Midjourney · Profitable; Series D, \$1B+ valuation estimates

V7 complete rebuild with improved coherence and speed; pioneering personalization efficiency (5 minutes vs. 200 images); video generation features under development.

Ideogram [STARTUP]^[93] — Ideogram AI (Toronto) · \$96.5M total over 2 rounds (incl. \$80M Series A, Feb 2024, Andreessen Horowitz)

Leading text-to-image model (best-in-class text rendering); its open-weight 4.0 release closes the gap between proprietary and open image models — a notable omission next to Midjourney/Runway/Stability.

Flags / unverified

■ OpenAI's Sora technical report lacks implementation details; no arxiv preprint published as of May 2026

■ Runway GWM-1 world model exhibits documented limitations: causal reasoning failure (effects before causes), object permanence issues, and success bias

■ DALL-E 3 being deprecated in favor of native GPT Image 1.5; migration required by May 12, 2026

■ Stability AI facing reported financial difficulties and organizational turnover since March 2024; Emad Mostaque departed as CEO

AlphaFold + computational protein design

Impact so far. AlphaFold and computational design have accelerated protein engineering from random selection to intentional computational design, enabling discovery of novel therapeutics, enzymes, and materials while achieving Nobel Prize recognition in 2024.

Watch for next. Multi-domain protein complex prediction, conformational dynamics modeling, and generalized enzyme design across diverse catalytic mechanisms reaching clinical translation within 2-3 years.

a) People behind it

Name	Role / Affiliation	Type	Prizes
Demis Hassabis ^[98]	CEO of Google DeepMind Google DeepMind; Isomorphic Labs (co-founder & part-time)	Biz	★ Nobel Prize in Chemistry 2024 ★ 2023 Albert Lasker Basic Medical Research Award
John Jumper ^[99]	Director of Google DeepMind Google DeepMind	Biz	★ Nobel Prize in Chemistry 2024 ★ 2023 Albert Lasker Basic Medical Research Award ★ Member of National Academy of Engineering 2026
David Baker ^[100]	Director of Institute for Protein Design; HHMI Investigator University of Washington; Howard Hughes Medical Institute	Acad.	★ Nobel Prize in Chemistry 2024 (half award)
Tristan Bepler ^[101]	Co-founder and CEO OpenProtein.AI	Biz	—
Joshua Meier ^[102]	Founder and CEO Chai Discovery	Biz	—
Markus Buehler ^[103]	Department Head of Materials Science and Engineering; Lead on VibeGen MIT	Acad.	—
Alexander Rives ^[104]	Co-founder & Chief Scientist, EvolutionaryScale EvolutionaryScale	Biz	—

Sub-ideas inside this breakthrough

Sub-idea	Description	Examples
AlphaFold2 — single-chain folding ^[105]	AlphaFold2 (2021) solved the 50-year protein-folding problem at near-experimental accuracy on CASP14, and the 200M+ structure database opened it up to every biology lab on earth.	AlphaFold2 · AlphaFold Protein Structure Database (EBI) · ColabFold
AlphaFold3 — multimodal complexes ^[106]	AlphaFold3 (2024) extended prediction beyond proteins to nucleic acids, ligands, ions, and post-translational modifications, making it directly useful for drug discovery and not just structural biology.	AlphaFold3 · Isomorphic Labs · Chai-1
De novo design with RFdiffusion ^[107]	Baker lab's RFdiffusion (and successors) generate brand-new proteins for a target binding site or function — flipping protein engineering from selection to direct design.	RFdiffusion / RFdiffusion2 · ProteinMPNN · ESM-IF
Enzyme design ^[108]	AI-designed enzymes for novel reactions (luciferase, plastic degradation, sustainable chemistry) reached single-shot success rates that took years of directed evolution before — now at all 41/41 test cases for RFdiffusion2.	IPD enzyme designs · Kemp eliminase · Plastic-degrading enzymes
Protein language models ^[109]	ESM-2 / ESM-3 (Meta / EvolutionaryScale) treat protein sequences like text, enabling zero-shot mutation effect prediction, sequence generation, and embedding-based search across the proteome.	ESM-2 · ESM-3 (EvolutionaryScale) · ProtT5

b) Current research frontier (last 12–18 months)

- 01.** AlphaFold 3 (published May 2024 in Nature) predicts structures of proteins, nucleic acids, ligands, ions and modified residues with substantially improved accuracy for protein-ligand and protein-nucleic acid interactions compared to specialized tools.^[106]
- 02.** RFdiffusion2 (published December 2025 in Nature Methods) generates enzyme active sites from functional group geometries, designing scaffolds for all 41 test cases vs. 16 for previous methods.^[108]
- 03.** D-I-TASSER (published 2026 in Nature Biotechnology) outperforms AlphaFold2 and AlphaFold3 on both single-domain and multidomain proteins through deep learning potentials combined with iterative threading.^[110]
- 04.** VibeGen at MIT (announced March 2026)^[103] designs proteins by their conformational dynamics and motion, not just static structure, using agentic AI for molecular mechanics.^[103]
- 05.** David Baker's lab received \$7 million from Washington Research Foundation (March 2026) to advance AI-enabled enzyme design with applications in medicine, technology, and sustainability.^[111]
- 06.** IsoDDE roughly doubles AlphaFold 3 accuracy on protein-ligand structures at <20% sequence identity and predicts novel cryptic binding sites from sequence alone (Isomorphic Labs).^[112]
- 07.** Isomorphic Labs closed a \$2.1B Series B on May 12, 2026 (Thrive Capital lead) to fund IsoDDE deployment and clinical pipeline acceleration — one of the largest private AI-drug-discovery financings on record.^[113]
- 08.** Core de novo protein design challenges (novel folds, assemblies, binders) are 'close to being solved' per Baker et al. Nature review; next 5-10 years focus shifts to functional nanomachines, switches, and catalysts.^[114]
- 09.** Baker Lab Nature papers (May 22, 2026) describe de novo quasisymmetric protein nanocages with expanded internal volume, opening a computationally designed delivery vehicle class for genetic medicines.^[115]
- 10.** NLP-style scaling laws do not reliably transfer to protein language models; data growth on UniRef yielded non-monotonic and sometimes worse variant-effect prediction, implying effective diversity (not token count) drives biology foundation models. [BACKFILL, arXiv Jul 2025, traction: dedicated Align blog + GitHub + active debate vs ProGen3's 'biology has scaling laws' claim]^[116]
- 11.** A new protein language model from EvolutionaryScale/CZ Biohub (Rives et al., bioRxiv Jun 2026) beats established methods on biomolecular complex/antibody-antigen structure prediction, designs nanomolar miniprotein and single-chain antibody binders via simple search, and maps 6.8B sequences with 1.1B predicted structures.^[117]
- 12.** Doudna lab used AlphaFold to discover VIPR, an RNA-guided gene-silencing system (~180 aa, ~1/8 of Cas9) in phages (bioRxiv, Yoon et al.).^[118]

c) Who could be next

Chai Discovery [STARTUP]^[119] — AI drug discovery startup · Series B: \$130M (Oak HC/FT, General Catalyst); Series A: \$70M (Menlo Ventures); Total ~\$231M

Raised \$130M Series B (December 2025) at \$1.3B valuation for AI foundation models predicting molecular interactions; announced Eli Lilly collaboration (January 2026) for biologic drug discovery.

Generate Biomedicines [STARTUP]^[120] — Clinical-stage protein design company · Multiple rounds; pre-IPO valuation undisclosed
Filed for Nasdaq IPO (2026) with 312 employees including 138 M.D./Ph.D.s; first Phase 3 patient dosed January 2026 for GB-0895 (anti-TSLP antibody); received FDA Fast Track designation.

Iambic Therapeutics [STARTUP]^[121] — AI-driven drug discovery platform · \$100M+ oversubscribed financing round (2026)
Announced \$1.7B potential value collaboration with Takeda (February 2026) for AI drug design; NeuralPLexer model claims to outperform AlphaFold for protein-ligand prediction.

OpenProtein.AI [STARTUP]^[101] — AI protein engineering platform · Seed/early stage (not disclosed)

Founded by Tristan Bepler (MIT PhD '20) and Tim Lu (MIT Associate Professor PhD '07); democratizing AI-driven protein design tools for biologists.

Scribe Therapeutics [STARTUP]^[122] — Scribe Therapeutics · Previously backed by Andreessen Horowitz; Series B disclosed
Couples engineered CRISPR enzymes with AI guide-RNA design (DeepXE) and allosteric epigenetic editors; ASGCT 2026 data show 10-100x off-target reduction.

Anthropic — Claude Science / in-house drug development [LAB]^[123] — Anthropic · Internal initiative (no separate round disclosed)

Frontier AI lab moving into AI-native drug discovery with an in-house pipeline; signals verticalization of AI-for-bio.

mRNA vaccines + LNP delivery

Impact so far. mRNA vaccines with LNP delivery have moved from basic science proof-of-concept (Karikó/Weissman 2005) to >13 billion COVID-19 doses and now expanding into personalized cancer vaccines with multi-year survival benefits.

Watch for next. Multi-neoantigen cancer vaccine Phase 3 readouts in 2026, in vivo CAR-T therapies entering pivotal trials, and broader mRNA platform expansion into infectious diseases (influenza, RSV, HIV).

a) People behind it

Name	Role / Affiliation	Type	Prizes
Katalin Karikó ^[124]	Professor of Translational Oncology and Immunology; former Senior Vice-President BioNTech University of Szeged (Hungary); University of Pennsylvania (adjunct); formerly BioNTech (2013-2022)	Acad.	★ Nobel Prize in Physiology or Medicine 2023
Drew Weissman ^[125]	Roberts Family Professor in Vaccine Research; Director of Penn Institute for RNA Innovation University of Pennsylvania, Perelman School of Medicine	Acad.	★ Nobel Prize in Physiology or Medicine 2023
Stéphane Bancel ^[126]	Chief Executive Officer Moderna Therapeutics	Biz	—
Uğur Sahin ^[127]	Chief Executive Officer (departing end of 2026); Co-founder BioNTech SE (planning exit); launching new mRNA company with Özlem Türeci	Biz	—
Pieter Cullis ^[128]	Professor of Biochemistry and Molecular Biology; Director Nanomedicines Research Group; Co-founder Acuitas Therapeutics University of British Columbia; Acuitas Therapeutics	Acad.	—

Sub-ideas inside this breakthrough

Sub-idea	Description	Examples
Pseudouridine-modified mRNA ^[124]	Karikó & Weissman's 2005 finding that pseudouridine substitution stops innate-immune sensing made mRNA therapy possible — every approved mRNA product, including the COVID-19 vaccines, uses it.	Pfizer/BioNTech BNT162b2 · Moderna mRNA-1273 · Penn Institute for RNA Innovation
Lipid nanoparticle (LNP) delivery ^[128]	Pieter Cullis's ionizable lipid nanoparticles are the actual delivery vehicle behind mRNA medicine; without LNPs, the RNA degrades before it reaches a cell.	Acuitas Therapeutics · Moderna proprietary LNPs · GenVoy LNP platform
Personalized cancer vaccines ^[129]	mRNA-4157/V940 (Moderna/Merck) plus pembrolizumab cut melanoma recurrence by ~49% at 5 years — the first durable evidence that personalized neoantigen mRNA vaccines work.	Moderna mRNA-4157 (intismeran) · BioNTech BNT122 (autogene cevumeran) · Genentech RO7198457
In vivo CAR-T via mRNA-LNP ^[130]	Targeted mRNA-LNPs that engineer CAR-T cells inside the patient — no apheresis, no preconditioning chemo — went from preclinical to clinical (Capstan/AbbVie's CPTX2309) for autoimmune disease.	Capstan Therapeutics CPTX2309 (acquired by AbbVie) · Tessera Therapeutics · Umoja
Self-amplifying & circular RNA ^[131]	Self-amplifying mRNA (saRNA) and circular RNA dramatically reduce dose and extend protein expression, enabling lower-dose vaccines and longer-acting therapies; Arcturus's saRNA COVID-19 vaccine is approved in Japan.	Arcturus ARCT-154 · Replicate Bioscience saRNA · Orna circular RNA
Reduced-reactogenicity LNP-mRNA ^[132]	Lipid nanoparticle-mRNA formulation engineered to reduce IL-1-mediated adverse reactions while preserving antibody response.	IL-1 identified as mediator of fever without affecting antibody response
OspA mRNA-LNP (Lyme) ^[133]	Nucleoside-modified mRNA-LNP vaccine targeting <i>Borrelia burgdorferi</i> outer surface protein A.	OspA ST1 mRNA-LNP elicited borreliacidal antibodies in mice

Sub-idea	Description	Examples
Personalized neoantigen mRNA cancer vaccines ^[134]	Patient-specific mRNA vaccines encoding tumor-derived neoantigens, paired with checkpoint inhibitors and standard-of-care chemo, now showing durable Phase 1 benefit and progressing to Phase 3.	Autogene cevumeran (BioNTech/Roche) ~90% 6-yr survival in pancreatic Phase 1 · mRNA-4157 (Moderna/Merck) in multi-indication Phase 3

b) Current research frontier (last 12–18 months)

01. Moderna and Merck's mRNA-4157 personalized cancer vaccine combined with Keytruda sustained 49% reduction in recurrence/death risk in melanoma at 5-year follow-up (announced January 2026).^[129]
02. In vivo CAR-T cell generation using biodegradable polymer nanoparticles delivering mRNA directly to T cells (published 2026 in Science) achieved B-cell depletion of 95% without external cell engineering.^[130]
03. BioNTech/Regeneron combination BNT111 showed statistically significant improvement in overall response rate in Phase 2 advanced melanoma trial vs. historical controls.^[135]
04. Karikó and Weissman's 2005 discovery of pseudouridine incorporation reducing immunogenicity revolutionized mRNA therapy; their work has contributed to >13 billion COVID-19 vaccine doses globally.^[124]
05. Circular mRNA CAR-T cells generated via PIE platform show superior antitumor efficacy vs. linear mRNA through enhanced cytotoxicity and reduced exhaustion (published 2025-2026).^[131]

c) Who could be next

Moderna/Merck cancer vaccine program [STARTUP]^[136] — Moderna Therapeutics & Merck · Funded by Moderna (\$10B+ annual R&D) and Merck partnership

Eight Phase 2/3 trials ongoing in melanoma, NSCLC, bladder cancer, renal cancer; Phase 3 readout expected 2026; personalized neoantigen approach shows durable long-term benefit.

Capstan Therapeutics (AbbVie acquisition) [STARTUP]^[137] — In vivo CAR-T therapy startup (acquired by AbbVie 2025) · Acquired by AbbVie for \$2.1B (all-cash, 2025)

Initiated Phase 1 trial for CPTX2309 (anti-CD19 in vivo CAR-T) for autoimmune disease (June 2025); first clinical translation of in vivo mRNA-based CAR-T nanoparticles.

U■ur ■ahin & Özlem Türeci's new mRNA company [STARTUP]^[138] — Next-generation mRNA platform startup · To be spun from BioNTech with proprietary technology transfer

Launching end of 2026 with 'disruptive' next-gen mRNA innovations; receiving mRNA tech from BioNTech; binding agreement expected by June 2026.

Flags / unverified

■ Updated intismeran autogene (mRNA-4157) KEYNOTE-942 long-term efficacy: new value = 5-year follow-up: 49% reduction in recurrence/death, 59% reduction in distant metastasis/death, ~53% OS benefit vs pembrolizumab monotherapy

■ Updated mRNA seasonal flu vaccine regulatory status: new value = FDA VRBPAC voted 9-0 (Jun 18, 2026) for Moderna's mRNA-1010 (mFluvisa) seasonal flu vaccine (full approval 50-64; accelerated approval 65+); FDA decision expected Aug 5, 2026

■ Updated personalized mRNA cancer vaccine — long-term efficacy: new value = 5-yr KEYNOTE-942: 49% relative reduction in recurrence/death, 68.8% vs 49.1% recurrence-free

GLP-1 agonists

Impact so far. GLP-1 agonists have revolutionized type 2 diabetes and obesity treatment, shifting paradigm from symptomatic glycemic control to weight loss-driven metabolic improvement, with >\$20B annual market value by 2026.

Watch for next. Retatrutide NDA filing and approval (late 2026-2027), broader expansion into cardiovascular and renal disease indications, and emergence of next-generation agonists with improved efficacy/tolerability profiles.

a) People behind it

Name	Role / Affiliation	Type	Prizes
Jens Holst ^[139]	Professor Emeritus; Discoverer of GLP-1 hormone University of Copenhagen	Acad.	—
Mads Krogsgaard Thomsen ^[140]	Chief Scientific Officer Novo Nordisk	Biz	—
Lilly Tirzepatide Development Team ^[141]	Vice President of Diabetes Development Eli Lilly and Company	Biz	—
Juan Pablo Gonzalez ^[142]	Chief Medical Officer (implied from retatrutide development) Eli Lilly and Company	Biz	—
Lars Reinhard ^[143]	R&D leadership; Mounjaro development Eli Lilly and Company	Biz	—
Ania M. Jastreboff ^[144]	Professor of Medicine; Director, Yale Y-Weight Obesity Research Center; lead investigator, retatrutide TRIUMPH-1 Yale University School of Medicine	Acad.	—

Sub-ideas inside this breakthrough

Sub-idea	Description	Examples
Semaglutide (mono GLP-1) ^[139]	Once-weekly semaglutide (Ozempic / Wegovy) delivered ~15% weight loss in obesity trials and showed ~20% MACE reduction in SELECT — the first obesity drug with hard cardiovascular outcome data.	Ozempic (T2D) · Wegovy (obesity) · Rybelsus (oral)
Tirzepatide (dual GIP/GLP-1) ^[145]	Lilly's tirzepatide (Mounjaro / Zepbound) hit ~22.5% weight loss at top dose in SURMOUNT-1, leapfrogging semaglutide and triggering the obesity-drug supply crisis of 2023–24.	Mounjaro (T2D) · Zepbound (obesity, OSA)
Retatrutide (triple agonist) ^[146]	Retatrutide (GIP/GLP-1/glucagon) showed ~24% weight loss in Phase 2 and ~71 lbs / 28% in early Phase 3 — likely setting the new ceiling and adding glucagon-driven energy expenditure to the mechanism.	Lilly retatrutide TRIUMPH program
Oral GLP-1s ^[147]	Oral semaglutide and Lilly's orforglipron are removing the injection barrier and could expand the market by an order of magnitude — manufacturable as small molecules without peptide-supply bottlenecks.	Rybelsus (oral semaglutide) · Lilly orforglipron · Pfizer danuglipron (discontinued)
Indication expansion ^[139]	GLP-1s now have approvals or strong RCT data in cardiovascular disease (SELECT), kidney disease (FLOW), MASH/MASLD, obstructive sleep apnea (SURMOUNT-OSA), and emerging signals in Alzheimer's and addiction.	SELECT (CV) · FLOW (CKD) · SURMOUNT-OSA · ESSENCE (MASH)
Amylin analogs as next-wave obesity therapeutics ^[148]	Amylin-receptor agonists (e.g. petrelintide) restore leptin sensitivity to induce satiety with markedly better GI tolerability than incretins, enabling co-formulation with GLP-1/GIP agonists like enicepatide.	petrelintide (Roche/Zealand) ZUPREME-1 Phase 2 · enicepatide (CT-388) GLP-1/GIP Phase 2 · petrelintide + enicepatide fixed-dose combo Phase 2 (mid-2026)

Sub-idea	Description	Examples
Amylin agonism and muscle-preserving obesity combinations ^[149]	Beyond GLP-1: selective amylin receptor agonists and combos engineered to preserve lean mass (e.g., GLP-1 + myostatin inhibition) became a dominant ADA 2026 theme, addressing the lean-mass-loss limitation of incretins.	Lilly eloralintide (selective amylin agonist) + tirzepatide · Novo CagriSema and amycretin/zenagantide · Roche petrelintide amylin analog · semaglutide + bimagrumab (myostatin) lowering lean-mass loss fraction

b) Current research frontier (last 12–18 months)

01. Retatrutide (GIP/GLP-1/glucagon triple agonist) showed 28.7% weight loss and A1C reductions up to 2.0% in Phase 3 TRANSCEND-T2D-1 trial (announced March 2026).^[150]
02. Retatrutide demonstrated 71.2 lbs average weight loss and substantial osteoarthritis pain relief in Phase 3 obesity trial (announced February 2026), with 7 additional Phase 3 trials expected to complete in 2026.^[146]
03. Zepbound (tirzepatide) received additional FDA approval for moderate-to-severe obstructive sleep apnea in obesity (December 2024), marking first medication approved specifically for this indication.^[151]
04. Semaglutide (Novo Nordisk) is a GLP-1 agonist with 94% homology to native GLP-1, modified with DPP-4 protease-resistant alanine substitution, used at lower doses for Ozempic/Victoza diabetes and higher doses for Wegovy/Saxenda weight loss.^[140]
05. Oral semaglutide approved by FDA as first GLP-1 pill for weight loss, expanding non-injection delivery options while maintaining efficacy of injectable formulations.^[147]
06. Triple-agonist retatrutide produces bariatric-surgery-class weight loss (30%+) in a pivotal Phase 3 trial without surgery, redefining the achievable benchmark for obesity pharmacotherapy.^[152]

c) Who could be next

Retatrutide clinical program (Eli Lilly) [STARTUP]^[141] — Eli Lilly and Company · Eli Lilly internal R&D (\$3B+ annual pharma R&D)

Triple agonist showing superior efficacy vs. dual agonists; Phase 3 trials ongoing in obesity, type 2 diabetes, osteoarthritis, sleep apnea, MASH; potential NDA filing expected late 2026.

Oral GLP-1 platform development (Novo Nordisk, Eli Lilly, others) [STARTUP]^[147] — Multiple pharmaceutical companies · Multiple companies with \$10B+ combined R&D spending

Oral semaglutide approved; multiple oral GLP-1/GIP agonists in development; expanding accessibility of GLP-1 therapy beyond injection-dependent populations.

Novo Nordisk semaglutide pipeline expansion [STARTUP]^[140] — Novo Nordisk A/S · Novo Nordisk annual R&D budget (\$1B+)

Expanding semaglutide beyond diabetes and obesity into cardiovascular, kidney disease, and MASH indications; LEADER, SUSTAIN, and FLOW trial data expanding label potential.

Roche obesity portfolio (petrelintide + enicepatide) [LAB]^[153] — Roche · Roche internal R&D (large pharma); petrelintide via Carmot acquisition

Major new entrant validating the amylin pathway; petrelintide Phase 2 data positioned as a core asset alongside enicepatide combinations at ADA 2026.

Structure Therapeutics [STARTUP]^[154] — Structure Therapeutics (Nasdaq: GPCR) · Public (Nasdaq: GPCR)

Leading oral small-molecule GLP-1 developer (aleniglipron); phase 3 planned Q3 2026 — competitive with oral offerings from Lilly/Novo.

Flags / unverified

■ Updated VK2735 oral 13-week weight loss: new value = 12.2% mean weight loss at 13 weeks (Phase 2, highest dose), 97% >=5%, 80% >=10% — ECO 2026 (May 12, 2026)

■ Updated retatrutide_phase3_topline: new value = TRIUMPH-1 met primary endpoints; 28.3% weight loss at 80 wks on 12 mg, 30.3% at 104 wks in BMI>=35 subgroup; data at ADA 2026 (announced May 21, 2026)

■ Updated Retatrutide (GIP/GLP-1/glucagon) TRIUMPH-1 Phase 3 obesity readout: new value = Up to 28.3% mean weight loss at 80 weeks (12 mg); 65.3% reached BMI <30; matches bariatric outcomes

■ Updated mazdutide (GCG/GLP-1) vs semaglutide head-to-head (DREAMS-3): new value = 48.0% of mazdutide vs 21.0% of semaglutide patients achieved HbA1c<7.0% plus ≥10% weight loss; full data presented ADA 2026

■ Updated survodutide SYNCHRONIZE-1 full Phase 3 efficacy: new value = Full data at ADA 2026: 16.6% mean weight loss at 76 weeks vs 3.2% placebo; up to 85.1% achieved ≥5% reduction

■ Updated retatrutide Phase 3 TRIUMPH-1 mean weight loss (12mg, 80wk): new value = 28.3% (30.3% at 104 weeks); 45.3% achieved ≥30% loss

■ Updated orforglipron (Foundayo) vs oral semaglutide head-to-head (ACHIEVE-3): new value = 57.1% greater relative A1C reduction, 73.6% greater relative weight loss vs oral sema 14mg

■ Updated berobenatide (monthly GLP-1) Phase 2b weight loss: new value = up to ~15.9% (VESPER-1, non-placebo-adjusted, 32wk); advancing to 10 Phase 3 studies

■ Updated oral small-molecule GLP-1 — phase 2b peer-reviewed: new value = Nature Medicine: up to 11.3% placebo-adjusted weight loss at 36 weeks (120 mg)

CRISPR therapies (Casgevy + base/prime editing)

Impact so far. Casgevy approval (December 2023) marked first FDA-approved CRISPR therapy; 23+ clinical trials now active with base/prime editing showing efficacy in rare genetic diseases and potential for broader indications.

Watch for next. Intellia's FDA approval filing 2026, Beam and Prime Medicine accelerated approvals late 2026-2027, expanded in vivo CRISPR programs, and next-generation base/prime editors with improved efficiency and safety profiles.

a) People behind it

Name	Role / Affiliation	Type	Prizes
Feng Zhang ^[155]	James and Patricia Poitras Professor of Neuroscience; Core Member Broad Institute MIT McGovern Institute; Broad Institute of MIT and Harvard	Acad.	★ National Inventors Hall of Fame inductee 2026
David Liu ^[156]	Richard Merkin Professor; Director Merkin Institute; HHMI Investigator Harvard University; Broad Institute of MIT and Harvard	Acad.	★ 2025 Breakthrough Prize in Life Sciences ★ 2026 Harvey Prize ★ 2026 Franklin Institute Bower Award
Editas Medicine/CRISPR Therapeutics Leadership ^[157]	CEO (Casgevy development) CRISPR Therapeutics; Vertex Pharmaceuticals (co-developer)	Biz	—
John Leonard ^[158]	Chief Executive Officer Intellia Therapeutics	Biz	—
Keith Gottesdiener ^[159]	Chief Executive Officer Prime Medicine	Biz	—

Sub-ideas inside this breakthrough

Sub-idea	Description	Examples
Ex vivo CRISPR/Cas9 (Casgevy) ^[160]	CRISPR Therapeutics + Vertex's Casgevy was the first FDA-approved CRISPR therapy (Dec 2023) — ex vivo edit of HSCs to reactivate fetal hemoglobin in sickle-cell disease and beta-thalassemia.	Casgevy (exa-cel) · CTX001
Base editing ^[161]	David Liu's base editors (cytosine and adenine) make single-letter DNA changes without double-strand breaks — Beam (BEAM-101 for sickle cell, BEAM-302 for AATD) and Verve (PCSK9 for cardiovascular) are the lead programs.	Beam BEAM-101 / -301 / -302 · Verve VERVE-102 · Tessera Gene Writers
Prime editing ^[162]	Liu lab's prime editing, the 'search-and-replace' of genome editing, achieved its first-ever human evidence in 2025 with PM359 in chronic granulomatous disease, on an FDA accelerated-approval path.	Prime Medicine PM359 (CGD) · PM577 · Twin-prime editing
In vivo CRISPR via LNP ^[158]	Intellia's NTLA-2002 (lonvoguran ziclumeran) for hereditary angioedema cut attacks ~87% in Phase 3 — the first Phase-3-positive in-body CRISPR therapy and a template for the rest of the genome.	Intellia NTLA-2002 · Intellia NTLA-2001 (ATTR) · Verve VERVE-102 (PCSK9)
Epigenome editing ^[163]	dCas9 fused to silencing or activating domains can switch genes off or on without cutting DNA — Tune Therapeutics, Chroma Medicine, and Epic Bio are running the first programs (Tune's TUNE-401 for HBV in trials).	Tune Therapeutics TUNE-401 · Chroma Medicine · Epic Bio
In vivo prime editing reaches the clinic ^[164]	Prime Medicine's PM577a, an LNP-formulated prime editor delivered IV to hepatocytes, is the first in vivo prime editing therapy authorized for human trials, correcting the ATP7B H1069Q mutation in Wilson Disease.	PM577a (ATP7B H1069Q, Wilson Disease) · PM577 follow-on targeting R778L allele (preclinical)

b) Current research frontier (last 12–18 months)

- 01.** Casgevy (exagamglogene autotemcel) FDA-approved December 8, 2023, for sickle cell disease in patients ≥ 12 years; uses CRISPR/Cas9 to induce fetal hemoglobin, eliminating vaso-occlusive crises in clinical trials.^[160]
- 02.** Intellia's lonvoguran ziclumeran (in vivo CRISPR therapy) reduced hereditary angioedema attacks by 87% vs. placebo in Phase 3 HAELO trial (announced April 27, 2026); rolling FDA BLA submission initiated.^[158]
- 03.** Beam Therapeutics' BEAM-302 (base editing for alpha-1 antitrypsin deficiency) dosed 29 patients with well-tolerated safety profile and no dose-limiting toxicities; FDA RMAT designation granted; optimal 60 mg dose selected for pivotal development.^[161]
- 04.** Prime Medicine's PM359 (prime editing ex vivo therapy for chronic granulomatous disease) shows first human safety/efficacy data with FDA 'plausible mechanism' pathway grant; company pursuing accelerated approval with potential 2026 alignment meeting.^[162]
- 05.** As of 2026, 23+ clinical trials are active using base editing or prime editing for leukemias, hypercholesterolemia, AATD, sickle cell disease, beta-thalassemia, and chronic granulomatous disease.^[165]
- 06.** Intellia's lonvo-z became the first in vivo CRISPR therapy to win a positive pivotal Phase 3 (HAELO), cutting hereditary angioedema attacks 87% vs placebo ($p < 0.0001$) with 62% attack-free over 6 months; NEJM-published, rolling BLA underway for 1H 2027 launch.^[166]
- 07.** Autolus' fast-off-rate CD19 CAR-T obe-cel showed 83% (5/6) remission (per standard lupus criteria) at target dose, with several complete/partial renal responses, in severe refractory lupus in the Phase 1 CARLYSLE trial, presented at EULAR 2026.^[167]
- 08.** Beam BEAM-304 (in vivo LNP adenine base editor for PKU/PAH) cleared FDA IND June 18 2026; preclinical normalization of plasma Phe.^[168]
- 09.** Dual CD19/BCMA CAR-T (Beijing GoBroad Boren) induced sustained remission in 11 refractory systemic-sclerosis patients (EULAR 2026).^[169]
- 10.** VERVE-102 (Verve/Lilly) in vivo base editing cut PCSK9 $\sim 88\%$ and LDL-C $\sim 62\%$ ($n=35$, Heart-2), durable >1 yr, no dose-limiting toxicities (NEJM).^[170]
- 11.** bioRxiv preprint: in vivo epigenome editing gave the most durable PCSK9 silencing vs nuclease/deaminase editing and siRNA (mRNA editors in Arcturus LUNAR LNPs, mice).^[171]

c) Who could be next

Prime Medicine (prime editing platform) [STARTUP]^[172] — Prime Medicine Inc. · Series B: \$200M+ (2022); backed by Jeff Bezos and others

First-ever prime editing human data (PM359 for CGD); pursuing accelerated approval pathway 2026; expects clinical data may support approval with FDA alignment meeting ongoing.

Verve Therapeutics (base editing for cardiovascular disease) [STARTUP]^[173] — Verve Therapeutics Inc. · Public company; venture-backed with multiple funding rounds

VERVE-102 Phase 1b data showed $\sim 50\%$ LDL reduction in highest-dose patients; shifted to improved LNP formulation after VERVE-101 liver enzyme abnormalities; targeting PCSK9 base editing.

Intellia Therapeutics (in vivo CRISPR for rare diseases) [STARTUP]^[174] — Intellia Therapeutics Inc. · Public company; multiple funding rounds and partnerships

First Phase 3 in vivo CRISPR data (hereditary angioedema); rolling FDA submission 2026; expects U.S. launch H1 2027 if approved; expanding to other rare genetic diseases.

Beam Therapeutics (base editing for genetic disease) [STARTUP]^[161] — Beam Therapeutics Inc. · Public company; founded by Liu, Barrangou, and others with \$300M+ raised

BEAM-302 advancing to pivotal development for AATD with 60 mg optimal dose; BEAM-301 (GSD-Ia) data expected 2026; RMAT designation supporting accelerated pathway.

EDIT-401 (LDLR upregulation, in vivo gene editing) [STARTUP]^[175] — Editas Medicine · Public (NASDAQ: EDIT)

Lead in vivo program targeting hyperlipidemia/HeFH; $\geq 90\%$ LDL-C lowering in NHPs; FIH expected 2026 with human PoC by year-end; positions Editas as direct competitor to Verve/Lilly in cardiovascular gene editing.

Autolus Therapeutics [STARTUP]^[167] — Autolus Therapeutics plc · Public (Nasdaq: AUTL)

Commercial-stage CAR-T company (obe-cel/Aucatzyl approved in B-ALL) extending its fast-off-rate CD19 platform into autoimmune disease (lupus), a fast-emerging high-value indication for cell therapy.

Flags / unverified

■ Updated Casgevy commercial adoption: new value = \$43M Q1 2026 revenue; 500+ patients initiated

■ Updated Ionvo-z regulatory status: new value = Rolling BLA initiated April 2026; submission completion targeted H2 2026; potential US launch H1 2027 (first-ever in vivo CRISPR product)

■ Updated Casgevy commercialization (Q1 2026): new value = \$43.0M Q1 2026 revenue; >500 patients initiated globally

■ Updated VERVE-102 Phase 1b Heart-2 readout: new value = Single-dose in vivo base edit: -62% LDL-C, -88% PCSK9 at top dose; NEJM publication May 25, 2026; Phase 2 H2 2026

Quantum error correction (logical qubits below threshold)

Impact so far. Crossed theoretical threshold for scalable quantum computing; demonstrated exponential error suppression on real hardware for the first time, moving from physics milestone to engineering milestone.

Watch for next. Fault-tolerant quantum computers with 10+ logical qubits demonstrating practical speedup (2025-2026); Riverlane's MegaQuOp (1 million error-free operations) target by 2026.

a) People behind it

Name	Role / Affiliation	Type	Prizes
Hartmut Neven ^[176]	Founder and Lead, Google Quantum AI; VP Engineering Google	Biz	★ TIME's 100 Most Influential People in AI (2025)
John Preskill ^[177]	Director, Institute for Quantum Information & Matter; Professor of Theoretical Physics California Institute of Technology (Caltech)	Acad.	★ Leading figure in quantum information theory; multiple honors for QEC research
Ilyas Khan ^[178]	Founder & Chief Product Officer; Vice-Chairman of Board Quantinuum	Biz	★ Founding Chairman, The Stephen Hawking Foundation (2019); Founding Chairman, Topos Institute
Barbara Terhal ^[179]	Professor of Quantum Information; Research Scientist QuTech & DIAM, Delft University of Technology (TU Delft)	Acad.	★ Leading researcher in quantum error correction theory and surface codes
Steve Brierley ^[180]	Founder and CEO Riverlane	Biz	★ Built world's first QEC chip (DD1); \$75M Series C funding lead (2024)
David Hayes ^[181]	Senior Quantum Physicist; Logical Qubit Research Lead Quantinuum	Biz	★ Co-author of Microsoft-Quantinuum 800x logical qubit achievement (2024)

Sub-ideas inside this breakthrough

Sub-idea	Description	Examples
Surface code below threshold ^[182]	Google Willow (Dec 2024, 105 qubits) was the first hardware to show that increasing the surface-code distance suppresses logical error exponentially — the long-promised threshold theorem realized in silicon.	Google Willow surface-code experiment
Logical qubits with virtualization ^[183]	Quantinuum + Microsoft demonstrated 4 logical qubits with 800x lower error than physical qubits using ion-trap hardware and run-time mid-circuit measurement, validating a different path to QEC.	Quantinuum H2 / Helios · Microsoft Azure Quantum
Bivariate bicycle / qLDPC codes ^[184]	IBM's bivariate bicycle codes (Nature 2024) need ~10x fewer physical qubits per logical qubit than the surface code, potentially shrinking the path to fault tolerance from millions of qubits to ~hundreds of thousands.	IBM Quantum (Heron, Loon) · Bivariate bicycle codes
Cat qubits & bosonic codes ^[185]	Encoding a logical qubit in the photons of a microwave cavity (Alice & Bob, AWS) intrinsically suppresses bit-flip errors, reducing the QEC overhead and offering an alternative to surface codes.	Alice & Bob cat qubits · AWS Ocelot · Yale circuit-QED
Real-time decoding hardware ^[186]	Riverlane's DD1 and Google's custom decoders move syndrome decoding into dedicated FPGA / ASIC, the missing piece for sustained logical operation — the 'CPU' for the quantum 'GPU'.	Riverlane DD1 · Google QEC decoder · QEC decoder competitions
Logical-qubit advantage on real applications (neutral atoms) ^[187]	First demonstration that logical qubits outperform physical qubits on an end-to-end real-world task (differential equations) on a neutral-atom processor at 99.4% gate fidelity across 1,000 equations.	Pasqal neutral-atom run, 1000 differential equations, logical beats physical by ~50% avg, up to 10x on hard cases

Sub-idea	Description	Examples
Barbell codes (qLDPC) ^[188]	A new family of quantum low-density parity-check error-correcting codes designed for IQM's superconducting Constellation processor, reducing logical error rates and physical-qubit overhead versus the surface code.	IQM barbell codes on the Constellation processor
Dual-rail superconducting qubits for fault tolerance ^[189]	Erasure-biased dual-rail encoding (photon-loss/leakage detected as erasures) in superconducting circuits, now pursued by D-Wave via its Quantum Circuits Inc. acquisition as a lower-overhead route to logical qubits, distinct from transmon surface-code approaches.	D-Wave gate-model roadmap: 17 physical qubits 2026 -> 181 by 2028 -> 10 logical 2030 -> 100 logical 2032 · Built on Quantum Circuits Inc. dual-rail tech (acquired Jan 2026)

b) Current research frontier (last 12–18 months)

01. Google Willow (December 2024): 105-qubit processor achieved surface code error suppression below threshold with distance-7 code at 0.143% per-cycle logical error rate; first demonstration of exponential error suppression with increasing code size^[182]
02. Microsoft-Quantinuum (April 2024): Demonstrated 4 logical qubits with error rates 800x lower than physical qubits; 14,000+ experiments without error using trapped-ion hardware with qubit virtualization^[183]
03. IBM (2024): Surface code distance-5 implementation with logical error suppression, decoding in 135 microseconds on laptop; bivariate bicycle codes show 27x speedup at 0.3% circuit-level error rates^[184]
04. Quantinuum Helios (2025): 3rd generation ion-trap quantum computer with 99.921% 2-qubit gate fidelity; first complete quantum chemistry simulation using QEC on real hardware^[190]
05. Riverlane DD1 QEC Chip (2024): Dedicated quantum error correction ASIC/FPGA achieving logical error rate of 1 in a trillion; published in Nature Electronics with end-to-end hardware validation^[186]
06. NVIDIA Ising open AI models accelerate QEC decoding/calibration with sub-microsecond latency, 2.5x speedup and 3x accuracy gains over classical baselines.^[191]
07. US federal industrial policy is now explicitly steering quantum hardware build-out at the foundry level — a structural shift from research grants to manufacturing capacity formation, with \$2B in CHIPS funding distributed across nine vendors on May 21, 2026.^[192]
08. Flatiron Institute CCQ + Boston University demonstrated (Science, May 21 2026) that a tensor-network algorithm, running on a laptop, can classically simulate the same 100-qubit highly-entangled dynamics problem D-Wave used to claim 'beyond-classical' computation with its 5,000-qubit Advantage2 annealer in March 2025 — resetting where the quantum-advantage line sits.^[193]
09. IBM disclosed \$10B five-year quantum push (May 28, 2026) spanning R&D, manufacturing, partnerships and the Andron foundry with DoC (\$1B CHIPS proposed); targets Starling fault-tolerant machine by 2029.^[194]
10. On a trapped-ion QCCD processor, two optimized QEC code constructions (12-qubit Knill-inspired and 16-qubit tesseract colour code) combined with scalable error detection achieve logical error rates 11x to 800x lower than physical-circuit baselines; now published in Nature (vol 654, pp 349-355, 2026).^[195]
11. Quantinuum 98-qubit Helios (Nature, June 17): 99.9975% 1q / 99.921% 2q fidelity, 99.967% SPAM; Sandia certified beyond-classical operation.^[196]
12. HPE's hybrid quantum-HPC platform (HPE Discover, June 15-18) integrates Quantinuum + IQM QPUs with Cray supercomputing for enterprise quantum-classical workflows.^[197]
13. IQM directional tile codes: up to 1,000x lower logical error rate vs surface code at ~30 physical qubits/logical, native iSWAP gates on existing hardware.^[198]

c) Who could be next

PsiQuantum [STARTUP]^[199] — PsiQuantum (photonic quantum computing) · Significant DARPA backing; private funding round details proprietary

DARPA US2QC program finalist (Feb 2025) pursuing surface codes on photonic qubits; broke ground on largest US quantum computing project at Illinois facility; targeting utility-scale quantum computer by 2033

Atom Computing [STARTUP]^[200] — Atom Computing (neutral-atom quantum computing) · Series B/C rounds; corporate backing from aerospace partners

Scaling neutral-atom systems for QEC; announced 1,000+ qubit demonstration targets; competing with Quantinuum (trapped ions) and Google (superconducting) on QEC roadmaps

Alice & Bob [STARTUP]^[185] — Alice & Bob (bosonic quantum computing) · \$50M+ Series B; European backing

Developing cat-qubits and bosonic codes as alternative to surface codes; co-hosted Fault Tolerance Summit (Les Houches 2024); raised \$50M+ Series B

Anderon [STARTUP]^[194] — IBM (spinout) + U.S. Department of Commerce · \$2B total (\$1B IBM cash + IP/assets; \$1B proposed CHIPS Act incentive)

First purpose-built US quantum foundry — provides shared, scaled 300mm fabrication for IBM, Quantinuum, Diraq, Microsoft and other modalities, addressing the wafer-supply bottleneck for fault-tolerant systems.

Flags / unverified

■ None. Primary sources verified: Google Nature paper (Dec 2024), Microsoft blog (Apr 2024), Quantinuum press releases (2024-2025), IBM Research (2024), Riverlane Nature Electronics publication (2024).

■ Updated Physical-to-logical qubit ratio (memory): new value = 2:1 (per-round error $\sim 1.3e-13$)

■ Updated largest_quantum_simulated_protein: new value = 12,635 atoms (trypsin); IBM Heron 156q

■ Updated ibm_cloud_quantum_milestone: new value = 10 years; 156q Heron at $\sim 1.17e-3$ two-qubit error

■ Updated IonQ–SkyWater acquisition status: new value = Shareholder-approved May 8, 2026; $\sim \$1.8B$; closing expected Q2-Q3 2026

■ Updated PsiQuantum US government support: new value = \$100M CHIPS LOI signed May 21, 2026 for fault-tolerant photonic component development

■ Updated QuEra fault-tolerant roadmap / cloud availability: new value = Libra: 256+ logical qubits, $1e-6$ logical error rate, $\sim 1M$ logical ops, on AWS Braket by 2028

Reusable rockets (Falcon 9 + Starship catch)

Impact so far. Demonstrated booster reusability at scale (Falcon 9) and tower-catch recovery capability for heavy-lift (Starship/Mechazilla), reducing per-flight costs by 50%+ and enabling rapid cadence launch operations.

Watch for next. Starship Ship catch on tower (targeted 2025-2026); suborbital relaunch within 24 hours; Blue Origin New Glenn cadence increases; Stoke Space and Rocket Lab first operational flights.

a) People behind it

Name	Role / Affiliation	Type	Prizes
Elon Musk ^[201]	Founder, CEO, and Chief Engineer SpaceX	Biz	★ TIME Person of the Year (2023); multiple recognitions for SpaceX achievements
Gwynne Shotwell ^[202]	President and Chief Operating Officer SpaceX	Biz	★ National Academy of Engineering member; recognized for commercial spaceflight advancement
Tim Dodd ^[203]	Lead Starship Systems Engineer (public spokesperson for Starship updates) SpaceX	Biz	★ NASA commendations for rapid iteration and testing protocols
Lars Blackmore ^[204]	Director of Entry, Descent, and Landing (EDL); Falcon 9 Booster Landing Algorithm Lead SpaceX	Biz	★ JPL/NASA recognition for guidance and control algorithms
Josh Boehm ^[205]	Director of Launch & Recovery Operations; Mechazilla Tower Systems Lead SpaceX	Biz	★ Internal recognition for tower catch system design and execution

Sub-ideas inside this breakthrough

Sub-idea	Description	Examples
Falcon 9 booster reuse ^[206]	SpaceX has flown more than 380 booster reflights with one booster reaching 22+ flights — turning rockets from single-use ammunition into commercial aircraft and pulling launch cost below \$2,000/kg.	Falcon 9 Block 5 · B1062 (22+ flights)
Mechazilla booster catch (Starship) ^[205]	Starship IFT-5 (Oct 2024) caught the Super Heavy booster mid-air on the launch tower's chopstick arms — replacing landing legs with the tower itself, the architectural choice that makes 24-hour reflight plausible.	Starship IFT-5 / IFT-7 catch · Mechazilla launch tower
Fully reusable upper stage ^[207]	The much harder problem — bringing the upper stage back through orbital reentry and reusing it — is what Starship's Ship is designed for, and what Stoke Space's Nova attempts with active heat-shield cooling.	SpaceX Starship Ship · Stoke Space Nova · Rocket Lab Neutron
Fairing recovery ^[208]	Falcon 9 fairing halves are now recovered and reflowed on essentially every mission (~100% rate over 300+ flights), removing what used to be a \$6M throwaway part per launch.	Falcon 9 fairing recovery (Doug / Mr. Steven)
Heavy-lift competition opens up ^[209]	Blue Origin's New Glenn (first booster reuse Apr 2026), Rocket Lab Neutron (debut targeted 2026), and Stoke Nova mean Falcon 9 / Starship will not be the only reusable launch options for the first time.	Blue Origin New Glenn · Rocket Lab Neutron · Stoke Space Nova

b) Current research frontier (last 12–18 months)

01. Falcon 9 Booster Reusability Record (May 2026): 384+ booster reflights with 100% success rate; single booster B1062 reached 22 flights by early 2026; turnaround as short as 3 weeks between flights^[206]

02. Starship IFT-5 (October 13, 2024): Historic first catch of Super Heavy booster with Mechazilla 'chopstick' arms at launch tower; booster rested safely on tower arms ~7 minutes after liftoff on first catch attempt^[205]

- 03.** Starship IFT-7 (January 16, 2025): Second successful Super Heavy booster catch achieved; booster 14 (B14) caught after 8+ minutes of flight; demonstrates repeatability of tower catch system^[210]
- 04.** SpaceX Launch Rate Dominance (2026): 55 launches (54 Falcon 9, 1 Falcon Heavy) through May 6, 2026; targeting 140-145 Falcon 9 launches by year-end; industry leading reusability infrastructure^[211]
- 05.** Falcon 9 Pricing & Reusability Economics (May 2026): Advertised launch cost \$74 million; enables cost-leadership vs. competitors; 100% fairing recovery success on 307+ missions through February 2025^[208]
- 06.** Starship V3 to debut mid-May 2026 with ~3x prior payload capacity (100+ MT LEO); Pad 2 first flight at Starbase.^[212]
- 07.** NASA CLPS Moon Base awards: Astrobotic \$297.9M, Firefly \$144.2M, Intuitive Machines \$148.3M for four lunar deliveries (~2028).^[213]
- 08.** SpaceX plans consumer Starlink mobile service (direct-to-device + possible terrestrial network), leveraging ~\$17B EchoStar spectrum (disclosed at IPO roadshow, Jun 26).^[214]

c) Who could be next

Blue Origin New Glenn [STARTUP]^[209] — Blue Origin (heavy-lift reusable rocket) · Internal Blue Origin funding; Jeff Bezos backed
Successfully reused booster for first time (April 19, 2026; second and third New Glenn flights); competing with Starship for heavy-lift market; achieving operational reusability milestone

Rocket Lab Neutron [STARTUP]^[215] — Rocket Lab (partially reusable medium-lift rocket) · Public company (NASDAQ: RKLK); \$500M+ operational runway

Targeting Q4 2026 debut after propellant tank failure setbacks; pursuing booster reusability for medium-lift class; complementary to Starship/New Glenn in launch market segmentation

Stoke Space Nova [STARTUP]^[207] — Stoke Space (fully reusable medium-lift rocket) · Venture-backed; Series C through development phase

100% reusable architecture; full-flow staged-combustion engines (Zenith, Andromeda); targeting early 2026 first flight; heat-shield reentry recovery approach distinct from SpaceX/Blue Origin

Rocket Lab Neutron [STARTUP]^[216] — Rocket Lab USA (RKLK) · Public (RKLK); \$2.2B+ backlog

Most credible Falcon 9 alternative for medium-lift reusable launch (~13t reusable / 15t expendable, 9 Archimedes engines); growing backlog signals real demand pull, not just announcements.

Flags / unverified

■ None. All data from official SpaceX statements, Blue Origin announcements, and third-party space tracking (Wikipedia/Spaceflight Now) verified through May 8, 2026.

■ Updated Falcon 9 cadence checkpoint: new value = 45 payloads deployed in single rideshare on May 2-3, 2026

■ Updated starship_v3_first_flight_status: new value = Flight 12 (first V3) scrubbed May 21, 2026 — hydraulic pin tower-arm issue; retargeted May 22, 2026 from Starbase Pad 2

■ Updated Starship V3 first flight: new value = Starship V3 + Raptor 3 + Pad 2 debut succeeded May 22, 2026; 20 mock Starlinks deployed, Ship reached Indian Ocean target

■ Updated Blue Origin New Glenn status: new value = Vehicle lost on pad during May 28, 2026 static fire; planned June 4 Kuiper launch delayed

■ Updated SpaceX total booster recoveries / single-drone-ship landings: new value = 619th total recovery; OCISLY reaches 200 catches (June 3, 2026)

■ Updated Blue Origin New Glenn return-to-flight target: new value = Before end of 2026 (per CEO Dave Limp, June 2026)

■ Updated Falcon 9 single-booster reuse record: new value = 35 flights (booster B1067), achieved 2026-06-07/08

■ Updated SpaceX corporate/financial status: new value = Public company (Nasdaq: SPCX) as of 2026-06-12; IPO priced \$135/share, closed ~\$161 (+19%), ~\$2.1T valuation at first close (largest IPO ever)

Renewables + battery cost collapse

Impact so far. Battery costs collapsed 80% since 2010 (\$1,200 to \$108/kWh); solar & wind LCOE below fossil gas; renewable capacity additions accelerating; grid storage emerging as competitive energy arbitrage asset class.

Watch for next. Sodium-ion parity with LFP cost (2026-2027); Form Energy iron-air deployment at utility scale (2025-2026); Tesla Megablock (20 MWh modular unit) launch late 2026; long-duration (12-100hr) storage cost-competitiveness crossing threshold.

a) People behind it

Name	Role / Affiliation	Type	Prizes
Jay Whitacre ^[217]	CEO, CTO, and Founder; Trustee Professor in Energy Stratus Materials (and Carnegie Mellon University)	Cross	★ Founder of Aquion Energy (2008); recognized for cathode innovations in battery chemistry
Tejas Patel ^[218]	Co-Founder and CEO Form Energy	Biz	★ MIT Technology Review Climate Tech Company to Watch (2023-2024)
Varun Sridhar ^[219]	Executive Lead, CATL Sodium-Ion Battery Division CATL (Contemporary Amperex Technology Co., Limited)	Biz	★ CATL leads global battery market; Chief Scientist announcement on Naxtra mass production (2026)
Wang Xiangming ^[220]	General Manager, BYD Battery Division BYD (Build Your Dreams)	Biz	★ BYD world leader in EV and battery production; global EV market leadership
Andrew Zito ^[221]	Executive Lead, Energy Business; Tesla Megapack Division Director Tesla Energy (Tesla, Inc.)	Biz	★ Leadership of fastest-growing grid battery deployment company; Q1 2026 record deployments
Albert Hache ^[222]	Chief Commercial Officer; Executive Leadership BloombergNEF (Bloomberg Finance L.P.)	Biz	★ Director of most-cited renewable cost benchmarking reports globally

Sub-ideas inside this breakthrough

Sub-idea	Description	Examples
Solar PV cost collapse ^[223]	Utility-scale solar LCOE fell to \$38–78/MWh (\$20–45/MWh with PTC), making solar the cheapest source of new electricity in most of the world for 10+ years running per Lazard's annual benchmark.	Lazard LCOE+ · BloombergNEF Solar Module Index
Lithium-ion battery price below \$100/kWh ^[222]	BloombergNEF's annual price survey logged \$108/kWh for packs and \$99/kWh for BEV packs in 2024 — the threshold at which a new EV reaches sticker parity with a comparable ICE vehicle without subsidies.	BNEF battery price survey
LFP chemistry takeover ^[220]	Lithium iron phosphate (LFP) has displaced NMC for most stationary storage and a growing share of EVs — cheaper, no cobalt, longer cycle life, the dominant chemistry in Tesla Megapacks and BYD vehicles.	CATL LFP · BYD Blade Battery · Tesla Megapack (LFP)
Sodium-ion mass production ^[219]	CATL Naxtra (~\$40/kWh cell, 2026 mass production) and BYD's 30 GWh sodium-ion line are commercializing a chemistry that needs no lithium, no cobalt, no nickel — important for grid storage and emerging-market EVs.	CATL Naxtra · BYD sodium-ion · HiNa Battery
Long-duration storage (iron-air) ^[224]	Form Energy's iron-air batteries (~\$250/kWh, 100-hour duration) target the multi-day grid-storage gap that lithium-ion is too expensive to fill — first utility deployments with Xcel and Georgia Power are live.	Form Energy iron-air · EnerVenue metal-hydrogen · ESS Inc iron flow
High-energy-density sodium-ion at GWh scale ^[225]	Sodium-ion cells engineered for 261 Wh/kg (high-energy), 162 Wh/kg with -50C discharge (power), and 180Ah/>20,000 cycle storage variant — closing the gap to LFP while using a non-lithium supply chain.	Gotion Gnascient (261 Wh/kg) · CATL Naxtra · BYD sodium-ion · HiNa Battery

b) Current research frontier (last 12–18 months)

- 01.** BloombergNEF Battery Price Collapse (June 2025): Lithium-ion pack prices fell 8% to record low of \$108/kWh; stationary storage at \$70/kWh (45% drop YoY); BEV packs at \$99/kWh (2nd year below \$100/kWh threshold)^[222]
- 02.** Solar & Wind LCOE Competitiveness (Lazard 2025): Utility-scale solar \$38-78/MWh; onshore wind \$37-86/MWh; with PTCs solar drops to \$20-45/MWh; wind and solar least-expensive generation 10th consecutive year^[223]
- 03.** CATL Sodium-Ion Battery (April 2026): Naxtra technology at \$40/kWh cell level (20% cheaper than LFP); mass production targeting 2026; operating range -40°C to 70°C; 10,000 cycle life with 90% retention at -40°C^[219]
- 04.** Form Energy Iron-Air Battery (2024-2025): Commercial-scale factory operational in West Virginia; utility deals with Xcel Energy, Southern Company, Georgia Power; target ~\$250/kWh energy (reservoir); 100-hour duration; first deployments Q4 2025^[224]
- 05.** Tesla Megapack Deployment Surge (Q1 2026): Record 10.4 GWh deployed in Q1 2026 (156% YoY growth); full-year 2025 deployments 31.4 GWh (2.1x 2023); Shanghai Megafactory producing 1 GWh+ annually by Q2 2025^[226]
- 06.** JinkoSolar: 34.82% certified perovskite-silicon tandem cell (SIMIT-certified), up from 34.76%, near LONGi's 34.85% record.^[227]
- 07.** HZB/Humboldt ESTI-confirmed 25.5% perovskite-CIGS tandem cell (1 cm²), new world record.^[228]

c) Who could be next

BYD Sodium-Ion [STARTUP]^[220] — BYD (battery and EV manufacturer) · Internal BYD funding; \$37B+ market cap; profitable EV/battery leader

30 GWh sodium-ion factory breakground (Jan 2024); Q3 2025 delivery timeline; targeting <70% of LFP cost long-term; competing with CATL for sodium-ion market leadership

Northvolt (now Lyten acquisition) [STARTUP]^[229] — Northvolt (Swedish battery manufacturer) / Lyten (US battery company acquirer) · Lyten-backed restructuring; previously \$5B+ valuation

Bankruptcy restructuring (Nov 2024, Mar 2025); Lyten acquisition of Northvolt assets including 16 GWh capacity; Northvolt Ett resuming production Q4 2025; European manufacturing recovery watch

Envision AESC / HiNa Battery Technology [STARTUP]^[230] — Envision AESC (joint venture); HiNa Battery Tech (Chinese sodium-ion leader) · Corporate backing (Envision Group, HiNa venture funding)

Sodium-ion alternatives gaining traction; HiNa targeting mass production 2025-2026; diversifying battery chemistry supply chains away from LFP dominance

Ford Energy [STARTUP]^[231] — Ford Motor Company · Internal Ford-funded subsidiary, leveraging existing Glendale KY plant capex

US-assembled LFP BESS at 20 GWh/year scale targeting utilities, data centers, and industrial customers; repurposes the Kentucky EV-battery plant from the dissolved BlueOval SK JV; Lisa Drake named president.

QuantumScape [STARTUP]^[232] — QuantumScape Corporation · Public (NYSE: QS); backed by Volkswagen

Leading solid-state lithium-metal battery developer; anodeless ceramic-separator cells promise higher energy density and faster charging. Two major OEMs (VW/PowerCo, Honda) now validate its platform toward commercialization.

JinkoSolar [PUBLIC_COMPANY]^[227] — JinkoSolar Holding · Public (NYSE: JKS)

One of the world's largest solar manufacturers and a perovskite-silicon tandem record contender (34.82% certified, June 2026); key player pushing tandem cells past the single-junction limit toward mass production.

Flags / unverified

■ Northvolt status in flux (bankruptcy restructuring as of May 2026); actual Form Energy commercial deployment timing contingent on Q4 2025 utility project completion. All other figures from BloombergNEF, Lazard, Tesla official reports, and industry tracking verified.

■ Updated Firm 24/7 solar+storage LCOE (best regions): new value = \$54-82/MWh

■ Updated CATL sodium-ion capacity: new value = 40 GWh/year sodium-ion expansion announced May 9, 2026 (\$735M)

■ Updated Global energy storage 2025 installs (BNEF): new value = 112 GW / 307 GWh installed in 2025; 158 GW / 459 GWh forecast for 2026

■ Updated CATL sodium-ion (Naxtra) production status: new value = Manufacturing bottlenecks solved; mass production confirmed for 2026 (May 30, 2026)

■ Updated QuantumScape OEM partners: new value = Honda R&D added as second major OEM (multi-year joint development on solid-state lithium-metal cells & manufacturing), 2026-06-18

Self-driving cars / robotaxis

Impact so far. Robotaxis crossed from pilot to scaled commercial reality between 2023–2025 — Waymo passing 4M paid driverless trips with peer-reviewed safety data, Tesla launching FSD-based robotaxis, and Aurora running fully driverless freight — finally validating a 15-year capital cycle.

Watch for next. Whether end-to-end neural-net autonomy (Tesla, Wayve) catches up with Waymo's HD-map / LiDAR stack on safety; expansion of Waymo One beyond 5–10 cities; Aurora's commercial trucking ramp; and whether Chinese AV scale (Apollo Go, Pony) fragments the global market.

a) People behind it

Name	Role / Affiliation	Type	Prizes
Tekedra Mawakana ^[233]	Co-CEO Waymo (Alphabet)	Biz	—
Dmitri Dolgov ^[233]	Co-CEO and CTO Waymo (Alphabet)	Biz	—
Ashok Elluswamy ^[234]	Director of Autopilot / FSD Tesla	Biz	—
Sebastian Thrun ^[235]	Founder; Stanford Professor (emeritus from CS faculty) Kitty Hawk / Stanford University; founded Google self-driving project (2009)	Cross	★ DARPA Grand Challenge winner 2005
Chris Urmson ^[236]	Co-founder and CEO Aurora Innovation; previously CTO of Google Self-Driving Car (now Waymo)	Cross	—
Raquel Urtasun ^[237]	Founder and CEO; Professor Waabi; University of Toronto	Cross	—

Sub-ideas inside this breakthrough

Sub-idea	Description	Examples
Waymo's mapped + LiDAR robotaxi stack ^[238]	HD maps + multi-sensor (LiDAR/radar/camera) + rules-and-learning hybrid is the only stack with peer-reviewed safety data at commercial scale and with paid driverless rides in 5+ US cities.	Waymo One (Phoenix, SF, LA, Austin, Atlanta) · Waymo Driver
End-to-end neural-net driving ^[239]	Replace hand-coded planners with a single learned model trained on video — Tesla FSD v12+, Wayve LINGO/GAIA, Comma.ai. Same scaling-laws logic as LLMs, applied to driving.	Tesla FSD v12+ · Wayve LINGO-2 / GAIA-2 · Comma.ai openpilot
Autonomous trucking ^[240]	Long-haul freight on highways is structurally easier than urban robotaxis; Aurora launched commercial driverless freight (Dallas–Houston) in May 2025, with Kodiak and Waabi following — a near-term economic prize.	Aurora Driver · Kodiak Robotics · Waabi Driver
Generative-AI simulation ^[241]	Neural-rendering simulators (Waabi World, NVIDIA Cosmos, Wayve GAIA-1) generate synthetic edge cases at scale — closing the rare-event data gap that pure on-road miles can't fill.	Waabi World · NVIDIA Cosmos · Wayve GAIA-1 · Applied Intuition
Chinese robotaxi at scale ^[242]	Baidu Apollo Go, Pony.ai, and WeRide run commercial driverless services across multiple Chinese cities; Apollo Go has reported 9M+ cumulative rides — a parallel deployment story to the US Waymo arc.	Baidu Apollo Go · Pony.ai · WeRide

b) Current research frontier (last 12–18 months)

01. Waymo One scaled paid driverless rides from a single Phoenix geofence (2020) to commercial operation across Phoenix, San Francisco, Los Angeles, Austin, and Atlanta, surpassing 4 million paid driverless trips by early 2025 and on track for 10M+ in 2026.^[238]

02. Tesla launched its initial robotaxi service in Austin in June 2025 using FSD-only / vision-only AI inference (no LiDAR / HD maps), the first major commercial alternative to Waymo's stack.^[243]

03. Waymo's safety data (Aug 2024 paper, 7M+ rider-only miles) reported 81% fewer airbag-deploying crashes and 78% fewer injury-causing crashes vs. a human-driven benchmark, the first peer-reviewed at-scale evidence that robotaxis can be safer than humans.^[244]

04. End-to-end neural-net driving (Tesla FSD v12+, Wayve, Comma.ai) replaced hand-coded planners with a single learned model trained on video, mirroring the LLM-style scaling-laws approach in the autonomy stack.^[239]

05. Chinese robotaxi operators (Baidu Apollo Go, Pony.ai, WeRide) are running commercial driverless services across Beijing, Wuhan, and Guangzhou; Apollo Go reported 9M+ cumulative rides by mid-2025.^[242]

06. Waymo's first purpose-built robotaxi (Zeekr-made Ojai, 6th-gen Driver with cheaper sensor stack) went live with public rides on May 28, 2026.^[245]

c) Who could be next

Wayve [STARTUP]^[246] — Wayve AI Ltd. · Series C, \$1.05B led by SoftBank (May 2024); also NVIDIA, Microsoft

Pioneering end-to-end embodied-AI autonomy; raised \$1.05B Series C in May 2024 led by SoftBank, with NVIDIA and Microsoft participating; partnerships with Uber and OEMs to ship as a software-only AV stack.

Waabi [STARTUP]^[247] — Waabi Innovation Inc. · Series B, \$200M led by Uber & Khosla (June 2024)

Raquel Urtasun's startup is building generative-AI simulation-first autonomous trucking; raised \$200M Series B (June 2024) led by Uber and Khosla, targeting fully driverless trucking deployment in 2026.

Aurora Innovation [STARTUP]^[240] — Aurora Innovation, Inc. (NASDAQ: AUR) · Public (NASDAQ: AUR); ~\$2B in cash reserves

Public company; launched the first fully driverless freight runs between Dallas and Houston in May 2025 — a leading indicator that long-haul autonomous trucking ships before passenger autonomy generalizes.

Applied Intuition [STARTUP]^[248] — Applied Intuition Inc. · Series F-equivalent, \$250M at \$15B valuation (2025)

Simulation and ADAS infrastructure provider used by most Western OEMs; raised \$250M at \$15B valuation (2025), positioning as the AWS of autonomy software.

Stellantis x Wayve x Uber L4 robotaxi partnership [STARTUP]^[249] — Stellantis / Wayve / Uber · Non-binding MoU; no financial terms disclosed

Three-way ecosystem (OEM platform + mapless embodied-AI driver + global ride network) to mass-produce and deploy L4 robotaxis without city-by-city mapping; signals convergence on Wayve's end-to-end approach.

Flags / unverified

■ Tesla robotaxi rollout used safety operators in Austin in 2025 — fully driverless status varies by report

■ Waymo cumulative ride numbers cited from company-published statistics; independent audit limited

■ Updated Waymo service area: new value = 1,400 sq mi across 11 cities (May 13, 2026)

■ Updated waymo_freeway_status: new value = All freeway rides paused; 4 Sun Belt cities paused (May 21, 2026)

■ Updated Pony.ai global robotaxi fleet size: new value = 1,700+ vehicles as of 2026-05-24 (+250 since 2025 annual report); WeRide ~1,000 domestic

■ Updated Tesla Robotaxi Austin status (geofence vs active fleet): new value = Entire Austin metro geofence as of June 3, 2026, but only ~20 active unsupervised vehicles (down from 25 late-April peak); Dallas 3, Houston 6; fleet ramp deferred to late 2026/early 2027 pending FSD v15

■ Updated Waymo consumer monetization model: new value = Waymo Premier, an invite-only \$29.99/mo subscription (priority pickups, 10% cash back, free cancellations), launched in SF, LA, Phoenix on Jun 11, 2026

■ Updated Waymo geographic expansion: new value = Registered Waymo Germany GmbH for EU robotaxi entry (June 25 2026)

Humanoid robotics

Impact so far. Humanoid robotics moved from research demos to low-volume industrial production between late 2024 and May 2026 — Tesla and Boston Dynamics in factories, Figure on the BMW line, 1X taking consumer pre-orders, with VLA models (RT-2, π 0, Helix) finally enabling task generalization beyond hand-coded behaviors.

Watch for next. Whether Helix-class full-body autonomy generalizes to unstructured homes (1X NEO, Figure 03), whether teleop data factories drop unit-economic costs below \$50/hour, and whether Tesla Optimus V3 actually hits 1M units/year — the inflection from industrial pilot to mainstream tooling.

a) People behind it

Name	Role / Affiliation	Type	Prizes
Sergey Levine ^[250]	Co-founder & Research Lead Physical Intelligence	Biz	—
Jitendra Malik ^[251]	Professor, Robotics Researcher UC Berkeley EECS	Acad.	—
Dieter Fox ^[252]	Senior Director, Robotics Research NVIDIA / Allen Institute for AI	Cross	—
Pieter Abbeel ^[253]	Robotics & RL researcher Amazon / UC Berkeley	Cross	—
Brett Adcock ^[254]	CEO & Founder Figure AI	Biz	—
Marc Raibert ^[255]	Founder Boston Dynamics; Boston Dynamics AI Institute	Biz	—

Sub-ideas inside this breakthrough

Sub-idea	Description	Examples
Vision-Language-Action (VLA) foundation models ^[256]	End-to-end neural policies that combine VLM backbones with robot action tokens — RT-2, RT-X / Open-X-Embodiment, Physical Intelligence π 0/ π 0.5, and Figure's Helix enable zero-shot generalization across embodiments and tasks.	Google DeepMind RT-2 · RT-X / Open X-Embodiment · Physical Intelligence π 0 / π 0.5 · Figure Helix 02
Tactile sensing & dexterous manipulation ^[257]	Multi-axis fingertip and palm sensors (3–5 mN sensitivity) fused with vision close the in-hand-manipulation gap — slip detection, contact-rich assembly, and delicate grasping become tractable.	Sanctuary AI Phoenix 75-DOF hands · Figure 03 fingertip + palm sensors · Toyota Punyo soft-skin · PaXini DexH13
Sim-to-real RL & whole-body control ^[258]	Domain-randomized RL trained in Isaac Lab/Sim plus hybrid imitation pipelines now transfer zero-shot to hardware, with whole-body controllers handling balance and locomotion across embodiments.	NVIDIA Isaac GR00T N1.6 · Boston Dynamics electric Atlas · Humanoid-Gym · Fourier GR-1
Data factories & teleoperation scaling ^[259]	Large-scale teleop pools (45–50 tutors per 100 robots) and at-home gig-economy recording have driven cost-per-tutor-hour down ~60% to ~\$118 in 2026, accelerating data-hungry VLA training.	Tutor Intelligence DF1 · Hyundai Robotics Metaplant · Scale AI / Encord teleop · DoorDash chore data
Form-factor convergence (26–32 DOF bipedal) ^[260]	The industry has converged on ~26–32 DOF bipedal designs with 16–22 DOF tendon-driven hands, enabling vertical integration and shipment scaling toward 90K+ units in 2026.	Tesla Optimus V3 · Boston Dynamics electric Atlas · Figure 03 · Unitree G1
MolmoAct2 (open VLA) ^[261]	Open-weight Vision-Language-Action foundation model for bimanual humanoid manipulation, trained on 720 h of teleoperation, with public weights, dataset, and eval code.	87.1% real-world DROID success with unseen objects · ~37x faster inference vs MolmoAct1 · Open weights + 720 h dataset released

Sub-idea	Description	Examples
Ambi Robotics + Pickle Robot inbound-logistics automation ^[262]	Integrated trailer-unload (Pickle) to AI palletize (AmbiStack) with zero human handoff at the dock door.	Pickle Robot trailer-unloading arms · AmbiStack AI stacking
RCM constraint-consistent dynamic control for surgical robots (ICRA 2026 Best Paper, Medical Robotics) ^[263]	Remote-center-of-motion constraint-consistent dynamic control; TUM / Haddadin lab.	ICRA 2026 Best Paper in Medical Robotics

b) Current research frontier (last 12–18 months)

- 01.** Figure's Helix 02 (Jan 2026) shows unified full-body autonomy from a single VLA — dishwasher loading, bottle uncapping, pill extraction, syringe dispensing — without task-specific training.^[264]
- 02.** Tesla Optimus V3 entered internal factory deployment in Jan 2026 with ~1,000 robots in production at Fremont and Gigafactory Texas; Tesla targets mass production at 1M units/year by Q3 2026.^[265]
- 03.** 1X NEO took 10,000 pre-orders in 5 days (Oct 2025); the Hayward, CA factory began full-scale production in Q1 2026 with consumer delivery targeted for late 2026.^[266]
- 04.** Unitree G1 launched at \$16K (April 2026) with 23 DOF; Unitree filed for a Shanghai IPO at ~\$610M (March 2026), aiming for 20K shipments in 2026 (vs ~5.5K in 2025).^[267]
- 05.** Boston Dynamics' electric Atlas entered commercial production in Jan 2026 with all 2026 units committed to Hyundai and Google DeepMind, supported by autonomous battery-swapping for continuous operation.^[268]
- 06.** DreamZero (NVIDIA GEAR, 14B World Action Model on a video-diffusion backbone) achieves >2x zero-shot generalization vs state-of-the-art VLAs in real-robot experiments, real-time 7Hz control, and few-shot embodiment transfer in 30 min of play data.^[269]
- 07.** GAM (arXiv:2606.17046, ETH/KAIST) repurposes a 3D geometry foundation model for contact-rich manipulation, beating VLA/world-action baselines on accuracy, robustness, speed and size.^[270]
- 08.** NavWM (arXiv:2606.24101, CAS/Beihang) unifies latent world reasoning, action prediction and controllable generation via 'latent world tokens' for visual navigation.^[271]
- 09.** NVIDIA Halos for Robotics: full-stack physical-AI safety system (IGX Thor + Halos OS + certification lab); Agility Robotics first adopter.^[272]

c) Who could be next

Figure AI [STARTUP]^[254] — Figure AI · Series C ~\$1.5B at \$39.5B valuation (Sep 2025); ~\$1.9B total raised

Now fully independent from OpenAI on its own Helix VLA stack; reported \$39.5B valuation (Sep 2025) and BMW production deal — a likely IPO candidate.

Unitree Robotics [STARTUP]^[267] — Unitree Robotics · Series C ~\$520M at \$5.5B (Feb 2026); IPO ~\$610M filing

Best price/performance leader globally (G1 at \$16K); Shanghai IPO filing March 2026 unlocks ~\$610M for 4× capacity expansion targeting 20K units in 2026.

Physical Intelligence [STARTUP]^[273] — Physical Intelligence (Pi) · Founded 2024 by Levine, Abbeel et al; venture-backed (specifics undisclosed)

The leading VLA foundation-model lab; $\pi 0.5$ (April 2026) shows emergent open-world generalization, positioned to license to OEMs across the industry.

Sanctuary AI [STARTUP]^[274] — Sanctuary AI · Series B ~\$50–100M (estimate); Magna strategic partnership

75-DOF hydraulic hands with industry-leading dexterity and a Magna automotive partnership for production deployment.

Figure 03 production scale-up [STARTUP]^[275] — Figure AI · Series C+; ~\$39B reported valuation

First humanoid OEM to credibly hit hourly production cadence with autonomous (non-teleop) locomotion on uneven terrain - a leading indicator of commercial humanoid roll-out.

ROBOTERA [STARTUP]^[276] — ROBOTERA (Beijing) · >\$200M (May 8, 2026) led by SF Group, HSG, IDG Capital

China's leading commercial humanoid; logistics product-market fit with China Post and SF Group; >95% of core components developed in-house

Unitree Robotics [STARTUP]^[277] — Unitree Robotics · Targeting ~\$6.2B valuation, raising ~4.2B yuan (~\$616M) via STAR Market IPO; backers include Meituan, Sequoia China, Tencent, Alibaba, Ant Group

First embodied-AI company cleared for a China A-share (STAR Market) IPO and chosen as the hardware base for NVIDIA's open Isaac GR00T Reference Humanoid; 2025 revenue 1.699B yuan at ~60% gross margins, but carries U.S. national-security/legislative risk.

NEURA Robotics [STARTUP]^[278] — NEURA Robotics GmbH · Series C of up to \$1.4B (led by Tether; Qualcomm, Amazon, NVIDIA, Bosch, Schaeffler, EIB)

One of the largest robotics/physical-AI rounds on record; positions a European player against US/China humanoid leaders with a shared-learning 'Neuraverse' and 4NE1 humanoid.

Theker [STARTUP]^[279] — Theker · \$85M Series A (led by CRV; Samsung, Aglaé Ventures)

Reconfigurable-form factory robots as an alternative to fixed humanoids; strategic backing from Samsung, LVMH's Aglaé, and Inditex signals manufacturing/retail pull.

XDOF [STARTUP]^[280] — XDOF (robot teleoperation data) · \$70M launch round — Thrive Capital, Spark Capital, a16z, Lux Capital, WndrCo

Fills the teleoperation-data-collection gap; open-sourced ABC-130K (130K bimanual trajectories); serves frontier-lab customers via GELLO-derived teleop hardware.

Pickle Robot [STARTUP]^[262] — Pickle Robot Co. · Venture-backed (see coverage)

Trailer-unloading manipulation for inbound logistics; core warehouse-logistics-robots player surfaced via the Ambi integration.

Flags / unverified

■ Tesla Optimus V3 specs and production timeline come largely from non-Tesla sources; treat with caution.

■ Figure's claims about full Helix capabilities are demo-tested, not peer-reviewed for unseen-task generalization.

■ Humanoid market projections (90K shipments 2026, 1.2M by 2030) are highly speculative.

■ Updated Figure Helix-02 longest autonomous run: new value = Full 8-hour autonomous factory shift at near-human throughput (~3 sec/package)

■ Updated Figure 03 production rate: new value = 1 robot/hour at BotQ; 350+ Gen-3 units; new System 0 perception-conditioned whole-body control

Fusion energy

Impact so far. Fusion went from public-science promise to active commercial track between 2022 and 2026: NIF demonstrated repeatable net energy gain, CFS broke ground on SPARC with HTS magnets, and Helion shipped a D-T plasma machine on the way to a 50 MW commercial PPA with Microsoft.

Watch for next. First private-fusion net-electricity-on-grid demonstrations (CFS SPARC Q>1 in 2027, Helion Orion 2028), AI plasma control becoming standard, and validation of tritium breeding fuel-cycle infrastructure at commercial scale.

a) People behind it

Name	Role / Affiliation	Type	Prizes
Omar Hurricane ^[281]	Chief Scientist, Inertial Confinement Fusion Program Lawrence Livermore National Laboratory	Gov	—
Bob Mumgaard ^[282]	Co-founder & CEO Commonwealth Fusion Systems	Biz	—
David Kirtley ^[283]	Co-founder & CEO Helion Energy	Biz	—
Michel Laberge ^[284]	Founder & CTO General Fusion	Biz	—
Pietro Barabaschi ^[285]	Director-General ITER Organization	Gov	—
Sam Altman ^[286]	Chairman of the Board Helion Energy (also CEO, OpenAI)	Cross	—
Bernard Looney ^[287]	Independent director, Type One Energy (former CEO, BP) Type One Energy	Biz	—

Sub-ideas inside this breakthrough

Sub-idea	Description	Examples
Tokamak (magnetic confinement) ^[288]	The most-developed fusion approach: a toroidal magnetic chamber confines a hot plasma. CFS SPARC, ITER, KSTAR/EAST, JT-60SA, and DIII-D are the major active programs.	CFS SPARC (HTS magnets) · ITER · KSTAR (100M°C, 48s record) · EAST (403s H-mode)
Inertial confinement (laser-driven) ^[289]	High-energy lasers compress a fusion fuel pellet to ignition. NIF reached repeatable net energy gain (8.6 MJ from 2.08 MJ in April 2025) — the first machine of any kind to do so.	NIF (LLNL) · Marvel Fusion (proton-boron) · Pacific Fusion
HTS (high-temperature superconductor) magnets ^[290]	Rare-earth barium copper oxide (REBCO) tape enables 20+ tesla magnetic fields, cutting tokamak size and cost dramatically — the enabling technology for compact commercial fusion machines.	CFS 20-tesla TF magnet · Tokamak Energy ST40 / Demo4 · MIT-CFS HTS collaboration
AI plasma control ^[291]	Deep RL controllers stabilize plasma in real time on DIII-D, KSTAR, and TCV; DeepMind's TORAX simulator (May 2024) is now part of CFS's planning loop and broadly adopted across the field.	DeepMind TORAX · DeepMind + EPFL TCV (Nature 2024) · DIII-D + KSTAR ELM-suppression AI
Alternative confinement concepts ^[286]	Field-reversed configuration (Helion, 150M°C D-T, Feb 2026), sheared-flow Z-pinch (Zap Energy, 1.6 GPa pressures), stellarators (Wendelstein 7-X, 43s record May 2025), and acoustically driven MTF (General Fusion).	Helion Polaris (FRC) · Zap Energy FuZE-3 (Z-pinch) · Wendelstein 7-X (stellarator) · General Fusion LM26 (MTF)

b) Current research frontier (last 12–18 months)

01. NIF achieved 10+ ignition shots with a peak yield of 8.6 MJ (April 2025) — establishing repeatable net energy gain and a target-gain trend above 4:1 for inertial confinement fusion.^[292]

- 02.** Commonwealth Fusion Systems installed the first of 18 toroidal-field HTS magnets on SPARC in Jan 2026; first plasma is targeted for 2026 with a $Q>1$ demonstration in 2027.^[293]
- 03.** Helion Polaris reached 150 million °C with deuterium-tritium fuel (Feb 2026) — the first privately-built D-T fusion machine; Orion commercial plant construction began Jul 2025 with a 50 MW Microsoft PPA targeting 2028.^[286]
- 04.** KSTAR set a 100 million °C, 48-second plasma record (2024) with a 300-second target by 2026; Wendelstein 7-X reached a 43-second high-fusion-product plasma in May 2025.^[294]
- 05.** AI/RL plasma control was demonstrated on DIII-D and KSTAR in 2024; DeepMind's TORAX simulator (May 2024) is now embedded in CFS's daily operational planning and adopted across the field.^[291]
- 06.** A desktop-scale (sub-5-inch) electrostatic/magnetic 'orbitrap' fusion device heated plasma to roughly 11 million degrees C (above 1 keV apparent ion temperature) on less than \$50M of investment, demonstrating a compact path toward fusion conditions.^[295]
- 07.** Helion received world-first regulatory licenses (WA radioactive-materials + air-emissions) for its Orion commercial fusion plant; enabled by NRC byproduct-material framework (ADVANCE Act).^[296]
- 08.** CFS published 5 peer-reviewed papers (Journal of Plasma Physics) on the ARC plant's physics basis for ~400 MW net electricity.^[297]

c) Who could be next

Commonwealth Fusion Systems (SPARC) [STARTUP]^[298] — CFS (MIT spinout) · \$3.0B total; Series B2 \$863M (Aug 2025) led by Nvidia, Google, Bill Gates

Largest private fusion effort: \$3B+ raised, magnet assembly underway, first plasma targeted 2026 and $Q>1$ in 2027 — closest to a credible commercial tokamak.

Helion Energy (Polaris / Orion) [STARTUP]^[299] — Helion Energy · Series E+ via Sam Altman, Khosla Ventures; Microsoft PPA May 2023

Only private firm with a firm grid PPA (Microsoft 50 MW by 2028); FRC achieved 150M°C D-T (2026); Orion commercial plant under construction.

TAE Technologies (Norm / Copernicus) [STARTUP]^[300] — TAE Technologies · \$1.3B+ equity raised; \$150M latest round (Chevron, Google, NEA)

Neutral-beam injection breakthrough (Apr 2025) cut machine complexity ~50%; Copernicus targeting net energy before 2030; planned TMTG-style merger at \$6B valuation (mid-2026).

Tokamak Energy (ST40 / Demo4) [STARTUP]^[290] — Tokamak Energy · Multiple rounds; UK government and private capital

Spherical tokamak architecture with 11.8-tesla HTS magnets; claims 30–40× smaller plasma volume vs LTS designs and a clear path to ST-E1 commercial pilot.

Thea Energy [STARTUP]^[301] — Thea Energy · \$100M (May 27, 2026); leads Prelude Ventures, US Innovative Technology, General Innovation Capital Partners

Planar-coil stellarator as a simpler-to-manufacture alternative to tokamaks; \$100M to build a demonstration prototype amid AI-data-center power demand.

Focused Energy [STARTUP]^[302] — Focused Energy GmbH (Germany / Austin, TX) · \$240M Series A (June 2026) led by RWE; ~\$300M private + \$200M grants total

Now one of the most heavily funded fusion startups; inertial-confinement laser fusion using direct-drive targets at high rep-rate, building a 'Lighthouse' demo at a former RWE nuclear site.

Type One Energy [STARTUP]^[287] — Type One Energy Group · Series B (closing 2026)

Leading stellarator-path fusion developer (Infinity One prototype at TVA Bull Run, grid power targeted ~2034); closing Series B, recruiting major-energy leadership (Bernard Looney) — a key non-tokamak competitor to track.

Flags / unverified

■ Sam Altman is Chairman of Helion's board, not founder (David Kirtley founded the company).

■ TAE merger details and valuation are reports of plans, not finalized transactions.

■ Updated zap_energy_strategy: new value = Hybrid fission-fusion company strategy; Zabrina Johal named CEO (May 5)

■ Updated US federal fusion commercialization strategy: new value = Finalized DOE Fusion S&T Roadmap published 2026-06-09 targeting commercial fusion in the mid-2030s, implemented via a new Office of Fusion

■ Updated Helion Energy funding & valuation: new value = \$465M Series G (2026-06-04), \$15.5B post-money valuation, total raised ~\$1.5B; led by Thrive Capital

Brain-computer interfaces

Impact so far. BCIs crossed from lab to early commercial reality between 2024 and 2026: Neuralink put 5+ patients on the Telepathy implant, Precision Neuroscience took home the first wireless-cortical-BCI 510(k), BrainGate hit 110 cpm bimanual typing, UCSF restored speech at 78 wpm, and Meta shipped a 20.9 wpm sEMG wristband to consumers.

Watch for next. The first BCI Pre-Market Approval (likely Synchron), Neuralink's first Blindsight implant, Paradromics speech-restoration pivotal results, and whether the non-invasive consumer category (Meta Neural Band, Kernel) closes meaningful ground on invasive bandwidth.

a) People behind it

Name	Role / Affiliation	Type	Prizes
Leigh Hochberg ^[303]	Director, Center for Neurotechnology and Neurorecovery; IDE sponsor-investigator, BrainGate Brown University / Massachusetts General Hospital	Cross	★ Sean M. Healey International Prize for Innovation in ALS (2024)
Edward Chang ^[304]	Director, Speech Neuroprosthesis Lab UCSF Department of Neurosurgery	Acad.	—
Jaimie Henderson ^[305]	Director, Stanford Neural Prosthetics Translational Lab; Co-PI, BrainGate2 Stanford School of Medicine	Acad.	—
Thomas Oxley ^[306]	Co-founder & CEO Synchron Inc.	Biz	—
Matt Angle ^[307]	Founder & CEO Paradromics	Biz	—
Benjamin Rapoport ^[308]	Co-founder & Chief Science Officer Precision Neuroscience	Biz	—

Sub-ideas inside this breakthrough

Sub-idea	Description	Examples
Intracortical microelectrode arrays (Utah arrays) ^[309]	High-density silicon arrays implanted directly into motor cortex (Blackrock NeuroPort) record from individual neurons — the gold standard for invasive BCIs since the early 2000s.	BrainGate clinical consortium · Blackrock NeuroPort · Stanford / Brown / MGH BrainGate2
Endovascular BCI (Stentrode) ^[306]	Synchron's stent-electrode is implanted via blood vessels — no open-brain surgery — trading some signal fidelity for dramatically lower surgical risk.	Synchron Stentrode · FDA COMMAND trial · Synchron + Apple Vision Pro integration
High-bandwidth thin-film cortical interfaces ^[308]	Flexible subdural electrode films (e.g. Precision Layer 7) put 1,024+ electrodes on a hair-thin sheet, with FDA 510(k) clearance and a Medtronic surgical-navigation partnership in 2025–26.	Precision Neuroscience Layer 7 · Mount Sinai 4,096-electrode demo · Medtronic + Precision partnership
Speech decoding & synthesis ^[310]	UCSF's Chang lab and Stanford have pushed motor-cortex speech decoders to 78 wpm with personalized voice synthesis; the BrainGate consortium hit 110 cpm bimanual typing.	UCSF Chang lab speech BCI · Stanford inner-speech decoder · BrainGate bimanual typing
Non-invasive consumer BCI (EMG / EEG / fNIRS) ^[311]	Wearable surface-EMG and EEG/fNIRS systems (Meta Neural Band, Kernel Flow, OpenBCI) are reaching consumer scale at meaningful — if lower — bandwidths than invasive devices.	Meta Neural Band sEMG wristband · Kernel Flow TD-fNIRS · OpenBCI
Intraoperative high-density micro-EECoG BCI ^[312]	Temporary high-density micro-electrocorticography arrays placed during routine awake craniotomy that yield same-day speech and cursor decoding, lowering the barrier to BCI use in surgical mapping and short-term clinical applications.	Johns Hopkins micro-EECoG study in 4 patients (May 2026) · Precision Neuroscience Layer 7 Cortical Interface

b) Current research frontier (last 12–18 months)

- 01.** Neuralink's PRIME trial has implanted 5+ patients across multiple sites by mid-2025; the second patient ('Alex') achieved CAD control and gaming within days, and Blindsight (visual prosthesis) received FDA breakthrough device designation (Sep 2024) with first-in-human implant planned for 2026.^[313]
- 02.** Paradromics received FDA IDE approval (Nov 2025) for the Connect-One speech-restoration trial with the Connexus BCI, claiming ~200+ bps information transfer (10–20× higher than BrainGate, 100–200× higher than Synchron) via thread arrays thinner than a human hair.^[314]
- 03.** INBRAIN Neuroelectronics completed enrollment for the world's first graphene-BCI clinical trial (April 2026) with zero device-related adverse events in the first cohort; 1,024-contact wireless rechargeable arrays.^[315]
- 04.** Precision Neuroscience's Layer 7 received FDA 510(k) clearance (April 2025) — the first wireless cortical BCI cleared for market — and is now in 30-day chronic trials at six U.S. medical centers; Medtronic partnership announced January 2026.^[308]
- 05.** BrainGate reported rapid bimanual typing (110 cpm, 1.6% WER) in two paralyzed patients (Nature Neuroscience, March 2026); the consortium raised \$200M Series D in late 2025 for FDA PMA pivotal trials.^[316]
- 06.** Neuralink developing surgical robot for precision brain access with automated procedures and high-volume implant production targeted for 2026.^[317]
- 07.** Synchron's Stentrode is the first BCI integrated with Apple's BCI HID profile as a native input category, enabling hands-free Apple-device control for paralysis patients.^[318]
- 08.** Neuralink is developing a next-generation surgical robot capable of implanting threads in any brain region, paired with a 2026 push to high-volume implant manufacturing.^[319]
- 09.** A UC Davis-led intracortical BCI (Nature Medicine, 15 June 2026) enabled a man with ALS to communicate independently at home via speech-to-text (averaging 56 words/min) and cursor control without researcher support over 3,800+ hours, demonstrating long-term self-administered use.^[320]
- 10.** Stanford/Nature (June 17): every motor-cortex region encodes whole-body movement (8 BCI participants, 45 movements), upending the motor homunculus and implying fewer implants for versatile BCI control.^[321]
- 11.** China (NMPA) approved Neuracle's NEO epidural BCI for commercial use — first invasive BCI approved outside clinical trials.^[322]

c) Who could be next

Neuralink Blindsight [STARTUP]^[323] — Neuralink · Reported \$1B+ private equity, Musk-led

First visual-prosthesis implant planned in 2026 — direct visual-cortex stimulation that bypasses damaged optics. A major expansion from motor-only BCIs.

Synchron pivotal trial [STARTUP]^[306] — Synchron Inc. · \$200M raise (Nov 2025) for pivotal trial execution

Preparing FDA Pre-Market Approval submission across 4+ U.S. sites; if approved, the first BCI PMA could define the regulatory gold standard for the field.

Paradromics Connexus [STARTUP]^[307] — Paradromics Inc. · Series-stage venture funding (specifics undisclosed)

Extreme bandwidth (~200+ bps), first FDA IDE for speech restoration as a primary endpoint, 3 clinical sites planned for early 2026.

INBRAIN Neuroelectronics [STARTUP]^[324] — INBRAIN (Barcelona) · Spanish/EU biotech funding; WEF recognition signals next round

Graphene-electrode material breakthrough with bi-directional 1,024-contact arrays; WEF Technology Pioneer 2025 and zero adverse events in first cohort.

Rice/Baylor BrainGate team [LAB]^[325] — Rice University & Baylor College of Medicine · Federal/clinical trial grants (BrainGate consortium)

First Texas-based BrainGate site; pairs Rice computational/engineering with Baylor clinical neuroscience to advance intracortical BCI for tetraplegia assistive robotics.

Neuracle Technology [STARTUP]^[322] — Neuracle Technology (with Tsinghua University) · Private (China)

Maker of NEO, the first invasive BCI approved for commercial use (China NMPA); direct competitor context for Neuralink/Synchron.

Flags / unverified

■ Paradromics' 200+ bps figure is a manufacturer claim; not yet independently verified in human use.

■ BCI regulatory pathway is fragmented across 510(k) / IDE / PMA tracks; no unified FDA BCI guidance exists yet.

■ Updated neuralink_surgical_robot_capability: new value = Next-gen R1 robot can implant in any brain region (motor + non-motor); threads through intact dura; real-time motion compensation

AI accelerators / advanced compute

Impact so far. GPU-dominated AI compute fractured into specialized lanes between 2024 and 2026 — wafer-scale (Cerebras), inference ASICs (Groq, Etched), photonic interconnect (Lightmatter), and per-hyperscaler custom silicon (TPU/Trainium/Maia/MTIA) — pulling 20–70% of opex out of training and inference.

Watch for next. Rubin (2H 2026) and the Cerebras OpenAI 750 MW deployment going live; Lightmatter Passage co-packaged optics in production; neuromorphic chips (Loihi 3, NorthPole) crossing the practical edge-AI threshold; whether NVIDIA-Groq integration cements inference dominance.

a) People behind it

Name	Role / Affiliation	Type	Prizes
Jensen Huang ^[326]	CEO & Co-founder NVIDIA	Biz	—
Andrew Feldman ^[327]	Co-founder & CEO Cerebras Systems	Biz	—
Jonathan Ross ^[328]	Founder Groq (acquired by NVIDIA Dec 2025)	Biz	—
Jim Keller ^[329]	CEO Tenstorrent	Biz	—
Norm Jouppi ^[330]	Engineering Fellow; lead architect, Google TPU Google	Biz	★ ACM-IEEE Eckert-Mauchly Award
Nick Harris ^[331]	Co-founder & CEO Lightmatter	Biz	—

Sub-ideas inside this breakthrough

Sub-idea	Description	Examples
GPU scaling (Hopper → Blackwell → Rubin) ^[332]	NVIDIA's flagship line keeps doubling effective AI throughput per generation: Blackwell (208B transistors, ~20 PFLOPS FP4 per B200) ships at scale in 2025, Rubin (3× Blackwell, 1.2 EFLOPS FP8 training) in 2H2026.	NVIDIA H100/H200 (Hopper) · NVIDIA B200 (Blackwell) · NVIDIA Rubin / Rubin Ultra
Wafer-scale processors ^[333]	Cerebras WSE-3 (5nm, 4 trillion transistors, 900K cores, 44 GB on-chip SRAM, 21 PB/s memory) eliminates chip-to-chip movement entirely — enabling single-chip training of multi-trillion-parameter models.	Cerebras WSE-3 · Cerebras CS-3 system · OpenAI 750 MW Cerebras deal
Inference-specialized ASICs ^[328]	Fixed-function silicon optimized for narrow inference workloads — Groq LPU's deterministic conveyor-belt architecture and Etched's transformer-only Sohu ASIC compete with GPUs on \$/token.	Groq LPU (acquired by NVIDIA) · Etched Sohu · MatX
Hyperscaler custom silicon ^[334]	Cloud platforms have abandoned GPU monoculture: Google Trillium / TPU v6, AWS Trainium 3, Meta MTIA 300/500, Microsoft Maia 200 each deliver hyperscaler-specific compute economics.	Google TPU v6 Trillium · AWS Trainium 3 · Meta MTIA 300/500 · Microsoft Maia 200
Photonic & neuromorphic compute ^[335]	Lightmatter's photonic processor and IBM NorthPole / Intel Loihi 3 neuromorphic chips break with the GPU paradigm — using photons or spiking neurons for orders-of-magnitude better energy per inference for the right workloads.	Lightmatter Envisi / Passage · IBM NorthPole · Intel Loihi 3
Google four-partner inference TPU supply chain ^[336]	Google diversifying custom AI silicon by adding Marvell (memory processing unit + next-gen inference TPU) alongside Broadcom and MediaTek, signaling hyperscaler push to commoditize inference accelerators and reduce NVIDIA dependence.	Marvell-designed MPU paired with TPU · Inference-only TPU separate from training-focused TPUs · ~2M MPU production target

Sub-idea	Description	Examples
NVIDIA RTX Spark (N1X) client AI superchip ^[337]	Arm-CPU + Blackwell-GPU SoC with 128GB unified memory bringing data-center-class local AI inference (120B-param LLMs, 1M-token context) to consumer Windows PCs.	RTX Spark Superchip / N1X · Windows-on-Arm agentic AI laptops from Dell/HP/Microsoft/ASUS/Lenovo/MSI

b) Current research frontier (last 12–18 months)

01. NVIDIA Rubin GPUs ship in 2H 2026 with ~3× Blackwell performance and 1.2 EFLOPS FP8 training capability; Rubin Ultra follows in 2027.^[326]
02. Cerebras Systems signed a >\$10B agreement with OpenAI (Jan 2026) for ~750 MW of inference compute through 2028 and filed for IPO at ~\$23B valuation in April 2026.^[338]
03. NVIDIA acquired Groq for ~\$20B (Dec 2025) and integrated the LPU into its 2026 GTC stack; Groq 3 LPU achieves ~1,500 tokens/sec on agentic AI inference.^[339]
04. Hyperscaler custom silicon scaled across all four majors: Google Trillium (4× LLM training perf vs v5e, 2× HBM bandwidth), AWS Trainium 3 (2.52 PFLOPS MXFP8, 144 GB HBM3e), Meta MTIA generations 4× HBM bandwidth, Microsoft Maia 200 (10 PFLOPS FP4).^[334]
05. Intel Loihi 3 (Jan 2026) delivers 8M neurons / 64B synapses at 4nm running at ~1.2W peak vs 300W+ for GPU equivalents — neuromorphic compute crossing the threshold for practical edge AI.^[340]
06. Cerebras' \$3.5B IPO at ~\$26.6B valuation, anchored by a \$20B+ OpenAI compute deal, validates wafer-scale silicon as a credible alternative to GPU dominance for AI inference.^[341]
07. OpenAI + Broadcom 'Jalapeño' reticle-sized inference ASIC, design-to-tape-out in ~9 months, perf/watt 'substantially better' than SoTA, gigawatt-scale with Microsoft from end-2026.^[342]
08. SK hynix sampling 12-high HBM4E (16 Gbps/pin, 48 GB/stack, >20% better power, +17% heat resistance vs HBM4) for next-gen AI accelerators.^[343]
09. NVIDIA fully liquid-cooled AI server (closed-loop, 45°C supply, near-zero water); est. >\$4M/yr savings at 50 MW.^[344]

c) Who could be next

Tenstorrent [STARTUP]^[345] — Tenstorrent · Series B (2024) led by Khosla Ventures and others

Jim Keller's open-source RISC-V + Metalium ASIC pursuing GPU alternatives for hyperscale clusters via 12×400 Gbps Ethernet scale-out; Blackhole successor scaling.

Rebellions [STARTUP]^[346] — Rebellions Inc. · Series C 2025, \$1.4B valuation; Arm + Samsung Ventures + Synopsys

Korean accelerator with 4nm UCle-Advanced quad-chiplet architecture (REBEL-Quad), 144 GB HBM3e, claiming ~3.2× tokens/W vs H200; Arm + Samsung backed.

FuriosaAI (RNGD next-gen) [STARTUP]^[347] — FuriosaAI · Rejected \$800M Meta acquisition; raised capital via LG and OpenAI partnerships

Korean TSMC 5nm AI ASIC; founder declined Meta's \$800M acquisition (March 2025) and is pursuing a 2027 IPO with LG and OpenAI partnerships.

Lightmatter (photonic compute) [STARTUP]^[335] — Lightmatter · Series D ~\$400M (2024); SoftBank Vision Fund, GV

Shipping the Envisé photonic processor and Passage L200/L20 co-packaged optics — the leading commercial bet on photonic interconnect for AI datacenters.

Flags / unverified

■ Etched Sohu's 500K tokens/sec figure is a manufacturer claim and the part remained unshipped at the time of writing.

■ FuriosaAI declined-acquisition story is widely reported but specifics are second-hand.

■ Updated `nvidia_quarterly_data_center_revenue`: new value = \$75.2B Q1 FY27 (May 20, 2026 print); Vera Rubin initial shipments Q3 2026; Q2 guide \$91B +/- 2%

■ Updated nvidia_capital_return_policy: new value = \$80B additional buyback authorization; quarterly dividend raised 25x to \$0.25/share; \$20B returned in Q1 FY27

■ Updated Cerebras post-IPO status: new value = CBRS at \$266.90 on May 27, 2026 (+10.4%); \$66B market cap; \$20B OpenAI capacity deal active

AI dev ecosystem & agent primitives (MCP, skills, agents, orchestration)

Impact so far. In roughly 18 months the AI-agent stack produced a set of durable, fast-standardizing developer primitives — MCP for connectivity, Agent Skills for portable expertise, subagents and orchestration apps for parallel work, and SDKs plus evals for governance — now backed by both major labs and a wave of well-funded startups.

Watch for next. Watch for consolidation around open standards (MCP and Skills now under foundation/shared governance), multi-agent 'post-IDE' orchestration workspaces becoming the default interface, and evals/agent governance maturing into a required production discipline.

a) People behind it

Name	Role / Affiliation	Type	Prizes
David Soria Parra ^[348]	Member of Technical Staff; co-creator of the Model Context Protocol (MCP) Anthropic	Biz	—
Justin Spahr-Summers ^[348]	Engineer; co-creator of the Model Context Protocol (MCP) Anthropic	Biz	—
Charlie Holtz ^[349]	Co-founder & CEO of Conductor (parallel coding-agent orchestration app); previously growth at Replicate Conductor (Melly Labs), YC S24	Biz	—
Jackson de Campos ^[349]	Co-founder of Conductor; 'the human interface to AI organizations' Conductor (Melly Labs), YC S24	Biz	—
Harrison Chase ^[350]	Founder & CEO of LangChain; creator of LangChain framework and the LangGraph agent-orchestration runtime LangChain	Biz	—
Amelia Wattenberger ^[351]	Product Lead for Intent (Augment Code agent-orchestration workspace); Partner, Sutter Hill Ventures; ex-GitHub Next Augment Code / Sutter Hill Ventures	Biz	—

Sub-ideas inside this breakthrough

Sub-idea	Description	Examples
Model Context Protocol (MCP) ^[13]	An open standard giving AI assistants a universal two-way way to connect to external data and tools, replacing per-integration connectors with one protocol.	Pre-built MCP servers (Drive, Slack, GitHub, Postgres) · Early adopters Block and Apollo · Donated to the Linux Foundation's Agentic AI Foundation (Dec 2025)
Agent Skills (Claude Skills / portable skills) ^[352]	Modular folders (a SKILL.md plus scripts and resources) that agents discover and load on demand via progressive disclosure to specialize at tasks.	PDF/Excel/PowerPoint/Word document skills · skill-creator skill · Open cross-platform standard at agentskills.io (Dec 2025)
Subagents / multi-agent orchestration in coding agents ^[353]	Specialized sub-agents that each run in their own context window with custom prompts, tools, and permissions, letting a primary agent delegate and preserve its main context.	Claude Code Explore/Plan/general-purpose subagents · Custom subagents in ~/.claude · Subagents in the Claude Agent SDK
Agent-orchestration desktop/cloud apps ^[351]	Standalone workspaces to run and coordinate multiple coding agents in parallel across isolated git worktrees, then review and merge their work from one interface.	Conductor (run a team of coding agents on your Mac) · Augment Code's Intent (coordinator/implementor/verifier teams) · Zed / Codex parallel agents
Agent SDKs / frameworks ^[354]	Libraries and runtimes for building, orchestrating, and deploying multi-agent workflows programmatically.	Claude Agent SDK · OpenAI Agents SDK (Responses API) · LangChain LangGraph

Sub-idea	Description	Examples
Context engineering & evals / agent governance ^[355]	An emerging discipline focused on curating what enters an agent's context window and on systematically evaluating, grading, and governing agent behavior for production reliability.	OpenAI Evals for Agents (trace grading, prompt optimization) · OpenAI Guardrails + Connector Registry · LangChain context-engineering framing
Agentic commerce via MCP + x402 ^[356]	Exposing financial accounts and pay-per-use data/compute to LLM agents through an MCP server plus the x402 micropayment standard, letting agents transact autonomously within user-set limits.	Coinbase for Agents MCP server usable in ChatGPT and Claude · x402 used to pay for premium research APIs and on-demand compute without login · x402 standard backed by AWS, Anthropic, Circle, Near

b) Current research frontier (last 12–18 months)

01. MCP went from an Anthropic experiment (announced Nov 25, 2024) to a cross-industry standard; in Dec 2025 Anthropic donated MCP to the Agentic AI Foundation, a Linux Foundation directed fund co-founded with Block and OpenAI.^[357]
02. Anthropic launched Agent Skills on Oct 16, 2025 across Claude apps, Claude Code, the Agent SDK and the API, then on Dec 18, 2025 published Skills as an open cross-platform standard (agentskills.io).^[352]
03. OpenAI released the Responses API and open-source Agents SDK in March 2025, then at DevDay (Oct 6, 2025) launched AgentKit — Agent Builder, ChatKit, Connector Registry and expanded Evals — to move agents from prototype to production.^[355]
04. Conductor, the parallel coding-agent desktop app (YC S24), reports use at companies including Linear, Vercel, Notion and Stripe and announced a Series A on Mar 30, 2026.^[349]
05. LangChain, maker of the LangGraph agent-orchestration runtime, reached unicorn status with a \$125M Series B at a \$1.25B valuation in Oct 2025, led by IVP with CapitalG and Sapphire Ventures.^[350]
06. OpenAI Daybreak releases GPT-5.5-Cyber (85.6% CyberGym) + Codex Security (30M+ commits / 30K repos scanned), with government Trusted Access partnerships.^[358]
07. Agon (arXiv:2606.24177) autonomous research orchestrator ran 444 loops across domains with no human-written experimental code.^[359]

c) Who could be next

Conductor [STARTUP]^[349] — Conductor (Melly Labs) · YC Summer 2024; Series A announced Mar 30, 2026 (amount not confirmed in primary source)

Pioneer of the 'run a team of coding agents on your Mac' category; parallel agents in isolated worktrees, adopted at Linear, Vercel, Notion, Stripe.

Augment Code (Intent) [STARTUP]^[351] — Augment Code · Augment backed by ~\$472M total (incl. Sutter Hill Ventures); Intent in public beta

Intent is a 'post-IDE' agent-orchestration workspace using coordinator/implementor/verifier agent teams over a shared living spec; bring-your-own-agent (Claude Code, Codex, OpenCode).

LangChain [STARTUP]^[350] — LangChain · \$125M Series B at \$1.25B valuation (Oct 2025), led by IVP

Most widely adopted open-source agent framework; LangGraph is a leading multi-agent orchestration/memory runtime and LangSmith leads agent evals/observability.

Transformer Lab [STARTUP]^[360] — Transformer Lab (lab.cloud) · Not disclosed (open-source platform; for-teams commercial tier)

Open-source MLOps/experimentation layer (compute orchestration, training, eval, quantization) with a teams tier; a credible primary-source research shop on efficiency/quantization to track for the edge/efficiency lane.

Flags / unverified

■ Conductor Series A dollar amount unverified against a primary source (YC page confirms the round exists, dated Mar 30, 2026, but not the amount).

■ MCP co-creator attributions sourced from the a16z 'MCP co-creator' podcast; the official Anthropic MCP announcement does not name individuals.

■ Updated MCP protocol + enterprise agent primitives status (mid-2026): new value = MCP statelessness in the 2026-07-28 spec RC (published May 2026); Anthropic adds enterprise-managed MCP connectors (Okta first) and self-hosted sandboxes + MCP tunnels for private MCP server access from managed agents

Five cross-cutting themes

1. Scaling laws are the through-line.

Every one of the AI breakthroughs (transformers, reasoning models, diffusion) explicitly relies on a scaling-law thesis. Battery and solar cost-down curves (Wright's law) are the same shape: roughly 20% cost reduction per doubling of cumulative volume. Reusable rocketry is the same curve in cost-per-kg-to-orbit.

2. Verifiable rewards close the loop.

Reasoning models train on tasks with checkable answers. AlphaFold is verified against crystal structures. QEC has a hard threshold. CRISPR and obesity Phase-3 trials hit pre-registered endpoints. Where the reward is dense and verifiable, RL — and capital — pour in.

3. Delivery is the new bottleneck in biology.

mRNA vaccines, in vivo CAR-T, in vivo CRISPR, and personalized cancer vaccines share one stack: a payload wrapped in a lipid nanoparticle. Whoever solves non-hepatocyte targeted delivery unlocks the rest of the genome.

4. Reusability and amortization beat raw novelty.

Falcon 9's moat is the same booster flying 22+ times. MoE LLMs reuse a small fraction of parameters per token. Surface-code logical qubits trade more physical qubits for arbitrarily long lifetime. Battery cycle life, not raw energy density, is what brought \$/kWh-cycle below grid parity.

5. The lab → spinout → IPO path is now standard.

AlphaFold launched Isomorphic Labs. Baker's IPD seeded Xaira / Generate / Outpace. Karikó & Weissman seeded BioNTech. Doudna & Charpentier seeded CRISPR Tx, Editas, Intellia. Liu's lab spawned Beam, Prime, and Editas. Sutton & Barto's RL textbook seeded an entire research lineage now valued at hundreds of billions.

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